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Non-Hermitian topological enhanced electromechanical impedance method for ultrasensitive damage detection☆

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ABSTRACT

Electromechanical impedance (EMI) sensing is widely used for structural health monitoring, but its sensitivity to weak or incipient damage is often limited because damage-induced spectral changes are typically small and distributed over a broadband response. To address the challenge, we propose a non-Hermitian topological (NHT)-enhanced EMI framework that converts small electromechanical impedance variations into pronounced frequency shifts of a boundary-localized topological mode. An NHT circuit is first introduced and analyzed to establish the origin of its boundary sensitivity. We then develop a discrete electromechanical model to show how structural damage is projected as an effective boundary perturbation to the NHT circuit, leading to a first-order prediction of the topological frequency shift. The proposed method is further validated in a cantilever beam model, where the NHT-enhanced configuration exhibits 10 times larger frequency shifts than the conventional EMI benchmark. Finally, by combining multiple NHT sensing channels with different operating frequencies and sensing locations, we build a machine learning model to achieve quantitative damage evaluation and localization in both beam and plate structures. Our findings establish a new avenue for ultrasensitive and data-driven structural health monitoring.

1. Introduction

With increasingly stringent requirements for safety, durability, and life-cycle management in modern engineering systems, structural health monitoring has become an essential component in the maintenance of civil, aerospace, and mechanical structures [1–5]. A central challenge lies in the detection of weak or incipient damage before it evolves into a critical defect. To address this need, a wide range of sensing techniques has been developed, including acoustic emission, guided waves, optical- or strain-based sensing, and vibration-based methods [6–10]. While these approaches have demonstrated broad usefulness, they often require relatively complex instrumentation, large-area measurement, or elaborate signal processing to reliably identify subtle damage.

Among existing sensing methods, piezoelectric electromechanical impedance (EMI) sensing has attracted sustained attention because it directly couples the dynamics of a host structure to an electrical quantity that can be measured in a compact and efficient manner [10–17]. When a piezoelectric transducer is bonded to a structure and electrically excited, the measured impedance reflects the coupled electromechanical response of the full system [18–25]. Since structural damage alters the local stiffness, mass, or damping characteristics, it also changes the measured EMI spectrum [26]. This mechanism makes EMI sensing attractive for structural health monitoring due to its simplicity and compatibility with integrated transducers.

However, the effectiveness of conventional EMI sensing fundamentally limited by how damage is encoded in the measured spectrum [27]. For weak or incipient damage, the resulting spectral variations are often small and distributed across multiple resonances rather than concentrated in a single prominent feature [28]. Consequently, damage identification typically relies on subtle peak shifts, spectral variations, or statistical indices evaluated over a finite frequency range [29,30]. The central challenge, therefore, is to transform weak electromechanical signals into a sharper and more discriminative spectral signature.

In parallel with developments in structural sensing, recent progress in non-Hermitian physics has opened new avenues for ultrasensitive measurement architectures [31–38]. A prominent approach is exceptional-point (EP) sensing, in which eigenvalues and eigenvectors coalesce at a spectral degeneracy, giving rise to frequency splitting in response to a small perturbation of magnitude (ε) [39–44]. Originally proposed in optical systems for single-particle detection and thermal monitoring [45], EP sensing leverages this degeneracy

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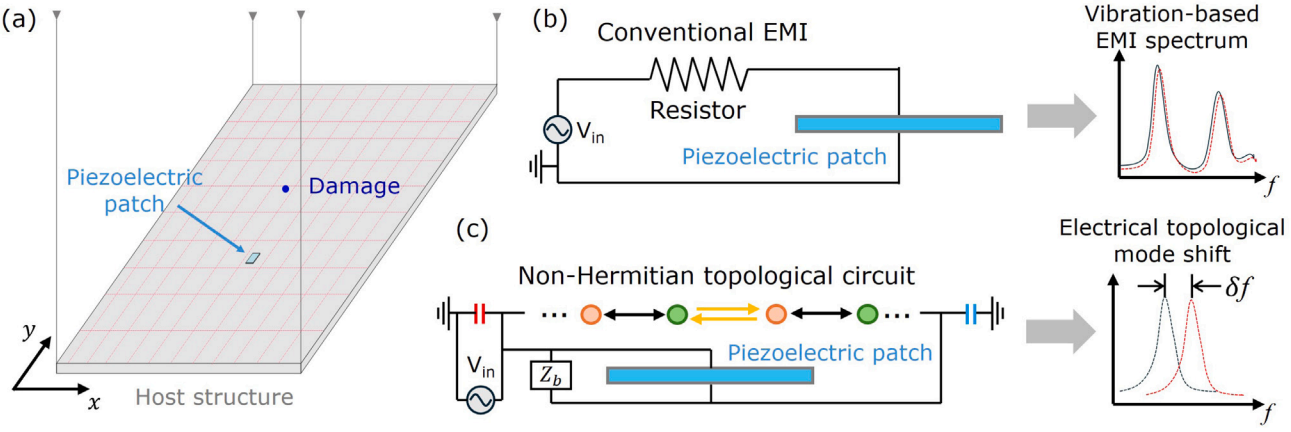


Fig. 1. Conceptual illustration of the NHT circuit-enhanced EMI method and a comparison with the conventional approach. (a) Schematic of the EMI setup: A piezoelectric patch is bonded to the host mechanical structure and connected to both an external circuit and a voltage source. (b) Configuration of the conventional EMI method. The black curve represents the undamaged EMI spectrum and the red curve represents the damaged EMI spectrum. (c) Configuration of the proposed NHT-enhanced EMI method. The black curve represents the undamaged topological frequency and the red curve represents the damaged topological frequency.

to achieve sensitivity far beyond conventional methods [46]. These ideas have since been extended to mechanical platforms, where structural damages or micro-masses induce eigenfrequency splitting at the EP. The first mechanical EP was proposed in an elastic beam with piezoelectric transducer and active feedback circuits [47], followed by successful demonstration of EP-based crack detection [48–50]. More recently, a high-order anti-P-pseudo-Hermitian mechanical system achieved third-order EPs through programmable nonreciprocal coupling, exhibiting multi-stage eigenvalue splitting and enhanced sensitivity to perturbations [51,52].

As an alternative to EP-based approaches, non-Hermitian topological (NHT) sensors provide a complementary method that exploits boundary-localized modes for robust, selective detection [53–56]. Quantitatively, EP-based sensing exhibits a frequency splitting $\Delta\omega_{\mathrm{EP}} \propto \epsilon^{1/p}$, where p is the order of coalescence [57,58]. However, this enhancement is realized only in the immediate vicinity of the EP. By contrast, NHT sensing operates away from EPs and relies on topological modes: when the measurand is mapped onto a boundary perturbation, the resulting topological frequency shift remains approximately linear in ϵ while exhibiting an exponential amplification with the number of sites in the NHT system ($\Delta\omega \propto \epsilon e^{N}$). For instance, Konye et al. developed an NHT ohmmeter, enhancing the accuracy of large resistance measurements [59]. Yuan et al. proposed NHT circuit sensors for ultrasensitive detection of distance, rotation angle, and liquid level [60]. This framework has advanced further with the realization of Floquet NHT sensors, which integrate time-periodic modulation and nonlinear feedback for dynamically enhanced sensitivity [61]. In addition, chip-scale NHT circuit sensors that utilize higher-order skin effects have achieved sub-femtofarad capacitance detection [62]. Despite these advances in electrical sensing, their potential for broader applications in structural health monitoring remains largely unexplored.

The NHT mechanism is well suited to EMI sensing because the electromechanical response of a bonded piezoelectric transducer can be incorporated into the circuit as an effective boundary perturbation [63,64]. On this basis, we propose an NHT-enhanced EMI framework for structural damage detection [65–67]. The central idea is to map damage-induced variations in the mechanical impedance of the host structure onto a boundary perturbation in an NHT circuit. NHT-enhanced EMI sensing converts weak damage-induced perturbations into a distinct frequency shift of an NHT mode. Building upon this concept, we integrate a non-Hermitian Su–Schrieffer–Heeger circuit with a piezoelectric transducer and investigate its sensing behavior through perturbation analysis and numerical simulations [68]. We demonstrate how small structural changes are translated into boundary electrical perturbations, how the resulting mode frequency shift depends on circuit size and operating conditions, and how this framework can be combined with machine learning to infer damage severity and location in beams and plates. In summary, this study establishes a physically transparent connection between conventional EMI sensing and non-Hermitian topological physics.

The remainder of this paper is organized as follows. In Section 2, we briefly introduce the physical concept and sensing mechanism of the proposed NHT-enhanced EMI method. Section 3 presents the design and theoretical formulation of the NHT circuit. Section 4 explains how variations in mechanical impedance are translated into boundary perturbations in the NHT circuit and how these perturbations give rise to the corresponding NHT frequency shifts. Section 5 demonstrates the sensing performance of the proposed method in a continuous structure. Section 6 presents machine-learning-assisted damage localization and evaluation for beams and plates. Section 7 provides the discussion and concludes the results.

2. Conceptual design of the NHT-enhanced EMI method

Fig. 1 summarizes the principles of the conventional EMI method and the proposed NHT-enhanced EMI method. In the conventional EMI approach, a piezoelectric patch is bonded to the host structure and connected to a simple external circuit, as shown in Fig. 1(a). Structural damage alters the local structural dynamics sensed by the patch, and this change appears as a variation in the measured EMI spectrum. In practice, however, weak damage often produces only small spectral shifts or slight line-shape variations, which can be difficult to identify robustly from the broadband response, as illustrated in Fig. 1(b). The proposed method uses the same piezoelectric interface and the same mechanical host, but replaces the conventional external circuit with an open boundary NHT circuit, as shown in Fig. 1(c). In this configuration, we do not directly read the electrical variation generated by the piezoelectric interface from the small distributed changes across the full EMI spectrum. Instead, this variation is introduced into the NHT circuit as an effective boundary perturbation. The sensing output is thus the frequency shift of a boundary-localized topological mode of the circuit.

This change of viewpoint is the key distinction between the conventional and proposed methods. The conventional EMI method relies on direct inspection of spectral changes in the piezoelectric impedance, whereas the NHT-enhanced EMI method converts the same electromechanical information into a single sharp frequency shift supported by the topological circuit. Because the topological mode is highly sensitive

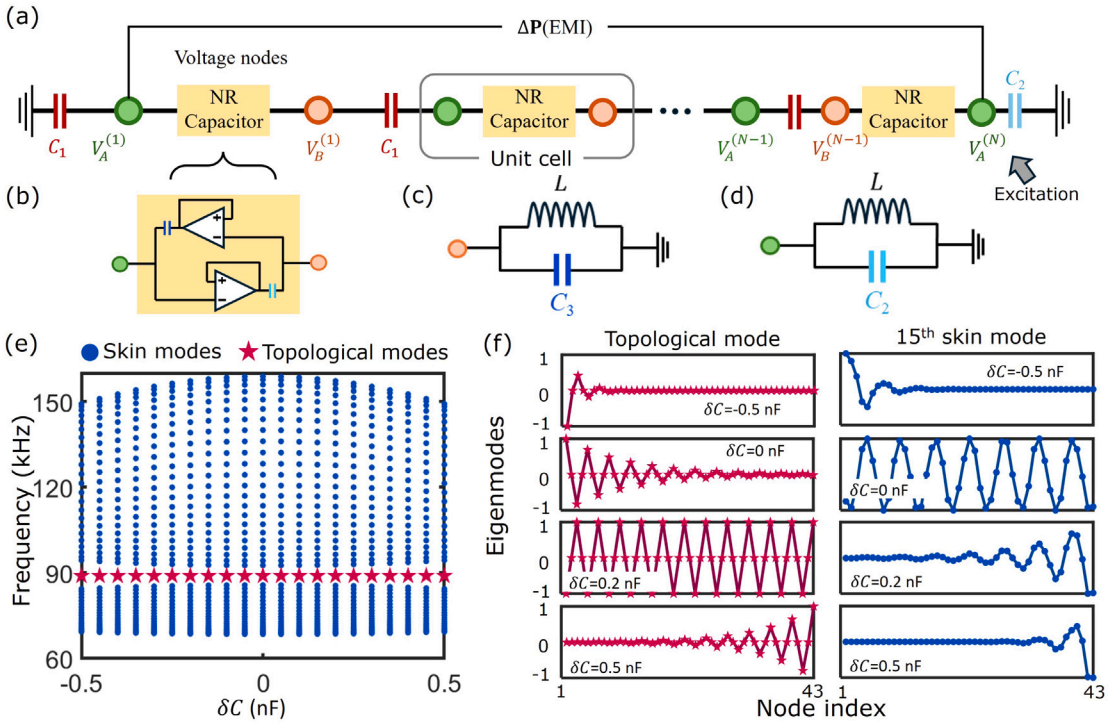


Fig. 2. The proposed non-Hermitian topological circuit and its modal analysis. (a) Schematic illustration of the circuit configuration. The measurand is connected to the two ends of the circuit. (b) Diagram of two unidirectional circuits for nonreciprocal coupling. (c) Grounded circuit of odd voltage nodes. (d) Grounded circuit of even voltage nodes. (e) Bulk band variation versus non-Hermitian strength. (f) Mode profiles of the in-band topological mode and the bulk mode under different non-Hermitian strengths.

to boundary perturbations, subtle structural changes that would otherwise be difficult to resolve in the conventional EMI spectrum can be transformed into a more distinguishable sensing signature.

To ensure that the boundary perturbation introduced by the piezoelectric interface remains within the sensitive operating range of the NHT circuit, a balancing impedance Z_b is employed to compensate the large static capacitive background of the patch. With this treatment, the NHT circuit primarily responds to the damage-dependent variation rather than the unperturbed capacitive offset. More details about Z_b can be found in Section 4.2 and Appendix H.

3. Theoretical formulation of the NHT circuit and its boundary sensitivity

In this section, we present the design principles of the NHT circuit and its boundary sensing mechanism. We begin by introducing the architecture of the NHT circuit and demonstrating its topological features under open boundary conditions [60,69–72]. Furthermore, we use perturbation theory to show how non-Hermitian topology enables exponentially amplified sensitivity to boundary impedance variations, laying the foundation for ultrasensitive structural health monitoring. The sensing limits and damping effects of the NHT circuit are also theoretically characterized. Finally, we simulate the temporal response of the topological mode to emulate practical measurement scenarios.

3.1. Design of the NHT circuit

NHT systems have recently emerged as a promising framework for ultrasensitive sensing. In particular, open boundary NHT chains exhibit exponentially enhanced sensitivity to boundary perturbations due to the non-Hermitian skin effect (NHSE). To apply this concept to structural health monitoring, we propose a discrete circuit network inspired by the non-Hermitian Su–Schrieffer–Heeger model, as illustrated in Fig. 2(a). Note that this is the unperturbed circuit network, that the effect of EMI is not yet included. Each unit cell contains two voltage nodes, $V_A^{(N)}$ and $V_B^{(N)}$, coupled through the intracell reciprocal capacitance C_1. Adjacent unit cells are coupled by nonreciprocal capacitors, as shown in Fig. 2(b). These nonreciprocal capacitors can be physically realized using directional operational-amplifier circuits. Each node is connected to a grounded impedance consisting of a reconfigurable inductor [73] and a capacitor [74] as shown in Fig. 2(c, d). The inductor is used to tune the topological frequency and the capacitor is used to preserve chiral symmetry. To support a non-Hermitian topological edge mode for sensing, the circuit must contain an odd number of voltage nodes. This asymmetry leaves one boundary node unpaired, allowing the emergence of a mid-gap state protected by chiral symmetry.

Applying Kirchhoff's current law to each voltage node in the frequency domain, the governing equations are written in the form of a quadratic eigenvalue problem:

$$\left[-\omega^2\mathbf{C}+\mathrm{i}\omega\mathbf{R}+\mathbf{L}\right]\mathbf{V}=0, \tag{1}$$

where $\mathbf{L},\mathbf{R},\mathbf{C}$ and $\mathbf{V}$ are inductor, resistor, capacitor and voltage matrices of the unperturbed NHT circuit, respectively. Details of these matrices can be found in Appendix A. The circuit parameters are set to $C_1 = 1.2$ nF, $C_2 = 1$ nF-δC, $C_3 = 1$ nF+δC, and $L = 1$ mH. For a circuit with N unit cells, the total number of voltage nodes is $N_e = 2N - 1$.

To better understand the modal behavior of the unperturbed NHT circuit, we analyze its frequency spectrum as a function of the nonreciprocal coupling strength $\delta C = (C_2 - C_3)/2$ by solving the quadratic eigenvalue problem in Eq. (1). Fig. 2(e) shows the eigenfrequencies under open boundary conditions (OBC). The red circles denote bulk (skin) modes, while the blue pentagrams indicate in-gap topological modes. As δC varies, the system undergoes a localization transition that alters the localization and the spectral separation of these modes. For small δC, the bulk modes remain extended or only weakly localized. However, when δC becomes sufficiently large, the eigenmodes exhibit strong asymmetric localization due to the NHSE. For the topological modes, the state is left-localized when the nonreciprocal coupling

strength is zero. As δC decreases, the topological mode remains left-localized, consistent with the skin effect. As δC increases, the topological mode first delocalizes at $\delta C = 0.25$ nF due to the skin effect and then relocalizes at the opposite boundary. Meanwhile, the topological frequency is not influenced by the nonreciprocal coupling strength. The exact frequency of the topological mode is

$$f_t = \frac{1}{2\pi\sqrt{LC_0}}, \tag{2}$$

where $C_0 = \sum_{i=1}^{3} C_i$. The designed NHT circuit therefore supports an in-gap topological mode with a fixed and predictable frequency, independent of the nonreciprocal coupling strength. This unperturbed topological frequency serves as a reference point for detecting perturbations introduced by structural damage.

3.2. Theoretical analysis of boundary perturbation on NHT circuit

To characterize the effect of boundary perturbations on the NHT circuit, we introduce an external impedance perturbation $\Delta\mathbf{P}$ applied across the two ends of the circuit. This boundary perturbation represents the effective EMI channel coupled to the NHT circuit, and can generally be expressed as an equivalent combination of capacitive and resistive contributions [75,76]. We write the boundary perturbation as

$$\Delta\mathbf{P}(\omega) = -\omega^2\Delta\mathbf{C} + \mathrm{i}\omega\Delta\mathbf{R}, \tag{3}$$

where $\Delta\mathbf{C}$ and $\Delta\mathbf{R}$ denote the effective perturbations in capacitance and resistance, respectively.

Using first-order perturbation theory for the quadratic eigenvalue problem, the eigenfrequency shift of both the bulk and topological modes induced by $\Delta\mathbf{P}$ is given by

$$\delta\omega = \frac{\mathbf{V}_L^{\top}\Delta\mathbf{P}(\omega)\mathbf{V}_R}{\mathbf{V}_L^{\top}(\partial\mathbf{P}/\partial\omega)\mathbf{V}_R}, \tag{4}$$

where $\mathbf{P}(\omega) = -\omega^2\mathbf{C} + \mathrm{i}\omega\mathbf{R} + \mathbf{L}$ and $\mathbf{V}_L$ and $\mathbf{V}_R$ are the normalized left and right eigenvectors of the unperturbed topological mode, satisfying the biorthogonal normalization condition $\mathbf{V}_L^{\top}\mathbf{V}_R = 1$ [77]. The right eigenvector also describes the modal voltage distribution of the NHT circuit.

For the in-gap topological mode [53], the left and right eigenvectors can be expressed as

$$\begin{aligned}\mathbf{V}_L &= a_l[1, 0, -\lambda_l, 0, (-\lambda_l)^2, 0, \ldots, 0, (-\lambda_l)^{\frac{N-1}{2}}]^{\top},\\ \mathbf{V}_R &= a_r[1, 0, -\lambda_r, 0, (-\lambda_r)^2, 0, \ldots, 0, (-\lambda_r)^{\frac{N-1}{2}}]^{\top},\end{aligned} \tag{5}$$

where $\lambda_l = C_2/C_1$, $\lambda_r = C_3/C_1$, and a_l and a_r are the normalization coefficients. The parameters λ_l and λ_r describe the spatial decay rates of the biorthogonal edge mode and therefore quantify its localization character. Specifically, when $C_3 > C_1$, the topological mode accumulates toward the right boundary; when $C_3 < C_1$, it localizes near the left boundary; and when $C_3 = C_1$, the mode becomes delocalized. These analytical features are consistent with the numerical mode profiles shown in Fig. 2(f).

To reveal the boundary-sensitive amplification mechanism, we first consider a purely capacitive boundary perturbation. In this case, the perturbation is introduced by an effective boundary capacitor C_p connected between the two ends of the circuit, so that

$$\delta\mathbf{C} = C_p\mathbf{B}, \tag{6}$$

where $\mathbf{B}$ is the boundary matrix, defined as $\mathbf{B} = \mathbf{b}_e\mathbf{b}_e^{\top}$. Here, $\mathbf{b}_e = \mathbf{e}_1 - \mathbf{e}_{N_e}$ and $\mathbf{e}_j$ is the jth canonical basis vector.

Combining Eqs. (4)–(6), the frequency shift of the topological mode is written as

$$\begin{aligned}\delta f =\ & 2\pi^2 f_t^3 L\, C_p a_l a_r\\ & \times \left[1 + (\lambda_l\lambda_r)^{(N-1)/2} - \lambda_r^{(N-1)/2} - \lambda_l^{(N-1)/2}\right],\end{aligned} \tag{7}$$

where f_t is the topological mode frequency. Detailed derivations are provided in Appendix A. For large N, Eq. (7) can be reduced to the asymptotic form

$$\delta f \approx C_p(\kappa_t\beta)e^{\alpha N}, \tag{8}$$

with $\kappa_t = \frac{2(1-\lambda_l\lambda_r)\pi^2 f_t^3 L}{1-(\lambda_l\lambda_r)^{(N+1)/2}}$ and

$$[\alpha, \beta] = \begin{cases}[(\ln\lambda_l)/2, \lambda_l^{-0.5}], & \text{if } \lambda_r < 1 < \lambda_l,\\ [(\ln\lambda_r)/2, \lambda_r^{-0.5}], & \text{if } \lambda_l < 1 < \lambda_r,\end{cases} \tag{9}$$

Eq. (8) shows that the topological frequency shift is approximately linear with respect to the perturbation magnitude C_p, while its dependence on the circuit size is governed by the exponent α. Therefore, α determines whether the NHT circuit exhibits size-amplified sensitivity.

This amplification exponent (α) is directly linked to the spectral winding of the periodic Bloch bands around the topological frequency. The Bloch matrix of the periodic system reduces to

$$\mathbf{P}(k, \omega) = -\omega^2\begin{pmatrix} 0 & C_1 + C_2e^{-ik}\\ C_1 + C_3e^{ik} & 0\end{pmatrix}, \tag{10}$$

and the winding number of the topological mode ω_t is defined as

$$\nu(\omega_t) = \int_0^{2\pi}\frac{dk}{2\pi i}\frac{d}{dk}\log[\omega(k) - \omega_t]. \tag{11}$$

Equivalently, because the off-diagonal part of the Bloch matrix controls the encircling behavior at ω_t, the winding can be evaluated from the two factors $C_1 + C_2e^{-ik}$ and $C_1 + C_3e^{ik}$, giving

$$\nu = \begin{cases}+1, & C_3 > C_1 > C_2,\\ -1, & C_2 > C_1 > C_3,\\ 0, & \text{otherwise.}\end{cases} \tag{12}$$

On the other hand, the exponent α in Eq. (8) can be written as

$$\alpha = \begin{cases}\frac{1}{2}\ln(C_3/C_1) > 0, & C_3 > C_1 > C_2,\\ \frac{1}{2}\ln(C_2/C_1) > 0, & C_2 > C_1 > C_3,\\ \frac{1}{2}\ln(\max(C_2, C_3)/C_1) \le 0, & \text{otherwise.}\end{cases} \tag{13}$$

Hence, the sensitivity exponent and the spectral winding satisfy the following direct correspondence:

$$\begin{cases}\alpha > 0 \iff |\nu| = 1, & \text{Sensitive,}\\ \alpha = 0 \iff \text{winding transition,} & \text{Insensitive,}\\ \alpha < 0 \iff \nu = 0, & \text{Insensitive.}\end{cases} \tag{14}$$

This result establishes the theoretical origin of the sensitive regime in the NHT circuit. When the winding is nonzero, the topological mode is accompanied by non-Hermitian skin effect, so the biorthogonal overlap at the two ends of the circuit grows with N, leading to the exponentially amplified relation $\delta f \propto e^{\alpha N}$. In contrast, when $\nu = 0$, the topological mode still responds linearly to the boundary perturbation, but this response no longer increases with circuit size [63]. Therefore, a nonzero winding number of the topological mode is the key condition for realizing size-enhanced sensitivity in the proposed NHT sensing mechanism.

3.3. Sensitivity analysis and the effect of damping at boundary

Fig. 3(a) summarizes four representative regimes of the proposed NHT circuit, distinguished by the winding number ν and the size-scaling exponent α. Regions A and C correspond to the size-leveraged sensitive regime, characterized by $\alpha > 0$ and $\nu \neq 0$, whereas region B corresponds to $\alpha < 0$ and $\nu = 0$. The boundary between these regimes gives the critical case $\alpha = 0$. When $\delta C = -0.5$ nF, the topological winding lies in Region A, as shown in Fig. 3(b). In this case, the topological mode is enclosed counterclockwise by the

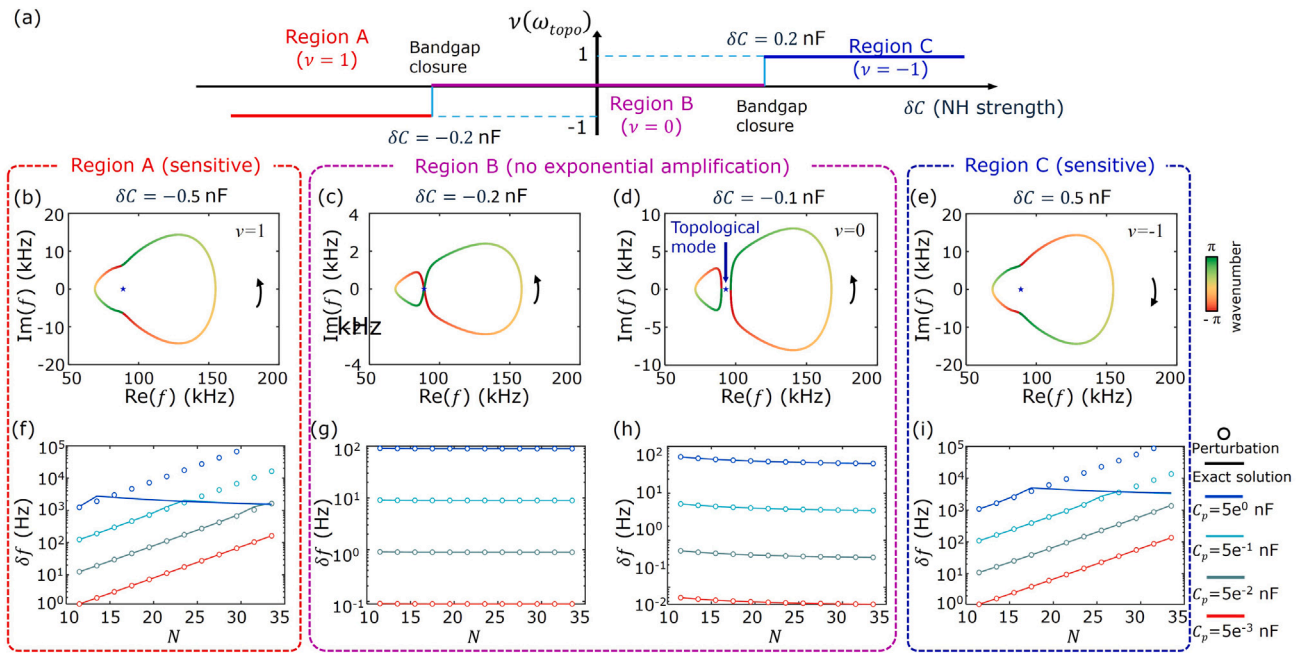


Fig. 3. The non-Hermitian topological properties of the proposed circuit and the sensing mechanism. (a) Winding number ν versus non-Hermitian coupling strength δC. (b–e) Winding curves and topological frequencies for (b) $\delta C = -0.5$ nF, and (c) $\delta C = -0.2$ nF, which is at the boundary of Regions A and B, and (d) $\delta C = -0.1$ nF, and (e) $\delta C = 0.5$ nF. (f–i) The frequency shifts of the topological mode corresponding to (b–e), respectively. The open circles represent the results from the perturbation method while the lines represent the exact solution of the eigenvalue problem.

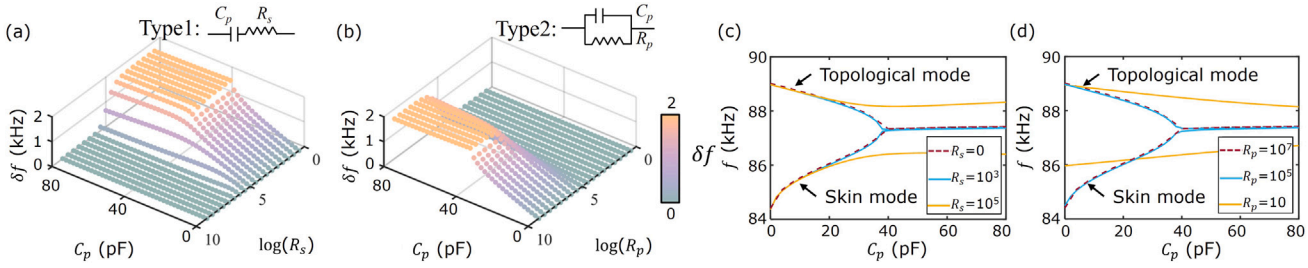


Fig. 4. Effect of damping on the sensing performance of the NHT circuit. (a) Frequency shift under different series-connected damping and perturbation strengths. (b) Frequency shift under different parallel-connected damping and perturbation strengths. (c) Frequency spectra of the topological mode and the nearest skin mode for series-connected resistors. (d) Frequency spectra of the topological mode and the nearest skin mode for parallel-connected resistors.

complex bands, corresponding to $\nu = 1$. From Fig. 3(f), we observe that, in this regime, δf increases approximately linearly with C_p, and is exponentially amplified as the circuit size N increases. The first-order perturbation results agree well with the solution of the exact eigenvalue problem for weak perturbations. As C_p further increases, the δf branch gradually approaches a sensing limit, beyond which it no longer follows the perturbation theory. This breakdown occurs because the topological mode does not remain spectrally isolated from nearby skin modes; instead, the shifted topological branch approaches a neighboring skin mode branch and eventually coalesces with it, forming a coupled mode [53]. The detailed coalescence process is presented in Appendix C. For $\delta C = -0.2$ nF and $\alpha = 0$, the complex band overlaps the topological frequency, indicating the critical winding transition, as shown in Fig. 3(c). Accordingly, Fig. 3(g) shows that δf remains approximately linear in C_p, while changing N has little influence on the frequency shift. When $\delta C = -0.1$ nF, the topological winding lies in Region B, as shown in Fig. 3(d, h). In this case, the topological mode is not enclosed by the complex bands and $\nu = 0$, so increasing the circuit size does not enhance the boundary sensitivity. Finally, Fig. 3(e) shows Region C, in which the topological mode is enclosed clockwise by the complex bands and $\nu = -1$. Although the winding direction is reversed, the circuit still satisfies $\alpha > 0$, so Δf remains approximately linear in C_p for small perturbations and is amplified with increasing circuit size N, as shown in Fig. 3(i). Fig. 3 reveals two important features of the NHT sensor: the winding number characterizes whether the topological mode is sensitive to boundary perturbations, while the sensitivity, or amplification ratio, is governed by α and remains valid only before coalescence with a skin mode.

In practical applications, EMI usually includes a real component, representing the effective resistance or damping of the EMI system [10]. Thus, to accurately model and predict the sensitivity of NHT circuits in realistic environments, it is essential to extend the boundary perturbation from a pure capacitor to a resistor–capacitor (RC) combination. We consider two types of boundary admittance perturbations: (i) a perturbed capacitor (C_p) in series with a resistor (R_s); and (ii) a perturbed capacitor (C_p) in parallel with a resistor (R_p). According to the general perturbation result in Eq. (4), the frequency shifts for these two cases, $\Delta\omega_1$ and $\Delta\omega_2$, can be written as

$$\Delta\omega_1 = -\frac{\mathbf{V}_L^T(-\omega_t^2 R_s C_p^2 \mathbf{B} + C_p \mathbf{B})\mathbf{V}_R}{\mathbf{V}_L^T(-2\omega_t R_s C_p^2 \mathbf{B})\mathbf{V}_R},$$
$$\Delta\omega_2 = -\frac{\mathbf{V}_L^T(\mathrm{i}\omega_t \mathbf{B}/R_p + C_p \mathbf{B})\mathbf{V}_R}{\mathbf{V}_L^T(\mathrm{i}\mathbf{B}/R_p)\mathbf{V}_R}. \tag{15}$$

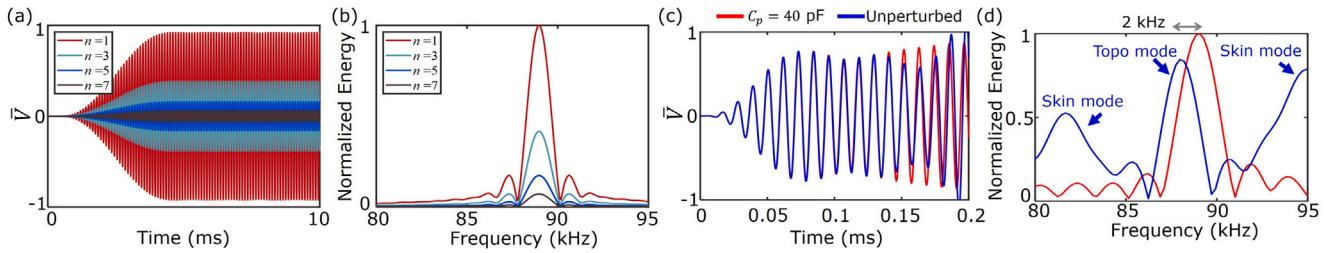


Fig. 5. Time domain simulation of the NHT circuit with and without perturbation. (a) Time domain signals of the 1st, 3rd, 5th, and 7th voltage nodes without perturbation. (b) FFT results of the time domain signals in (a). (c) Time domain signal at the 1st node with a perturbation of $C_p = 40$ pF. (d) Comparison of the FFT results at the 1st node with and without perturbation.

Figs. 4(a–b) illustrate how these perturbations affect the frequency shift δf of the topological mode. Here, the circuit is fixed at $\delta C = -0.5$ nF and $N = 23$. In the series case (Type 1), when $R_s \ll 1/(\omega C_p)$, the sensitivity is approximately the same as that of a pure capacitor. However, as R_s increases beyond 10^5 Ω, the factor $\omega_t R_s C_p \gg 1$ becomes dominant, thereby suppressing the capacitive contribution to $\Delta\omega_1$. In the parallel case (Type 2), when $R_p \ll 1/(\omega C_p)$, the resistive shunt dominates and behaves as a static load, suppressing the frequency shift even when C_p is large. By contrast, when $R_p > 10^6$ Ω, the damping effect becomes negligible, the perturbation again approaches purely capacitive behavior, and the sensitivity is recovered. The sensing limit is evident in Figs. 4(a, b), as indicated by the saturation of δf at large C_p. Moreover, increasing the series resistance or decreasing the parallel resistance reduces the sensing range. Figs. 4(c) and 4(d) show the formation of this sensing limit from the eigenfrequency spectra. We observe that the limit arises from the coalescence of the topological mode with a skin mode. In the series-connected, zero-damping case shown in Fig. 4(c), the topological mode frequency decreases while the skin mode frequency increases as C_p grows. Their frequencies change rapidly until they coalesce. Beyond this point, the resulting coupled mode becomes insensitive to further boundary perturbations, which explains the saturation observed in Figs. 3 and 4. When a resistor is introduced, damping makes the coalescence imperfect and causes the coalescence to occur at a smaller perturbation strength. As a result, the allowable perturbation range before the onset of saturation is reduced, leading to a narrower sensing range. Accordingly, the sensing range decreases with increasing R_s. Conversely, in the parallel case, the sensing range decreases with decreasing R_p.

3.4. Temporal response of the NHT circuit

Fig. 5 numerically demonstrates the NHT mode and its frequency shift in the time domain to emulate a real-world measurement scenario. In this case, $\delta C = -0.5$ nF, corresponding to a topological mode localized at the left boundary, *i.e.*, at the 1st node. A narrowband tone-burst signal with a central frequency $f_c = 1/(2\pi\sqrt{LC_0})$ is applied at the end of the circuit ($N = 23$). Fig. 5(a) shows the normalized temporal voltages $\bar{V}$ at the 1st, 3rd, 5th, and 7th voltage nodes. The steady-state voltage amplitudes are strongly localized near the boundary, indicating the presence of a localized topological mode, and the FFT results in Fig. 5(b) confirm the same spatially localized energy distribution. Further, a boundary perturbation $C_p = 40$ pF is applied across the two ends of the circuit. Fig. 5(c) compares the temporal voltage magnitudes at the 1st node with and without the perturbation, revealing a clear steady-state phase difference. The peak in the FFT results exhibits a frequency shift δf, as shown in Fig. 5(d), which confirms the perturbation-induced shift of the NHT mode.

4. Electromechanical discrete model of the NHT-enhanced EMI method

Although the sensitivity of the NHT circuit to an externally introduced boundary capacitance has been analyzed, the electromechanical origin of such a perturbation and its coupling to the host structure have not yet been quantitatively established. To address this gap, we next introduce a discrete electromechanical model to illustrate how structural variations modify the mechanical impedance and how these variations are translated into effective boundary perturbations in the NHT circuit, thereby shifting the topological frequency.

4.1. Electromechanical discrete model

As shown in Fig. 6, the host structure is modeled as a finite chain of 11 identical lumped masses m_0, connected by nearest-neighbor springs of stiffness k_0. The two end masses are further connected to ground by springs of the same stiffness k_0, which defines the intact reference state in the absence of damage. Each mass is also subjected to grounded viscous damping c_0. A piezoelectric transducer is bonded between the ith and $(i+1)$th masses, so that the electromechanical coupling is governed by their relative displacement. Here, we define the piezoelectric coupling location vector as $\mathbf{b}_m = \mathbf{e}_i - \mathbf{e}_{i+1}$. The electromechanically coupled deformation is then given by $\Delta u = \mathbf{b}_m^\top \mathbf{u}$, where $\mathbf{u} = [u_1, u_2, \ldots, u_{11}]^\top$ is the displacement vector of the masses.

Using the linear stress-charge constitutive relation, the coupled electromechanical equations can be expressed as

$$f_p = k_p \Delta u - \theta V_p, \qquad Q_p = \theta \Delta u + C_p^T V_p, \tag{16}$$

where k_p denotes the piezoelectric stiffness, θ is the electromechanical coupling coefficient, C_p^T is the clamped capacitance of the piezoelectric element, V_p is the voltage across its electrodes, and Q_p is the corresponding free charge. The resulting mechanical force exerted by the piezoelectric element on the spring-mass chain is written as $\mathbf{f}_p = f_p \mathbf{b}_m$.

We consider two types of damage: a mass perturbation δm applied to the jth mass and a stiffness perturbation δk applied to the spring connecting the $(j+1)$th and jth masses. Since the purpose of this discrete model is to examine weak damage sensitivity, we assume that $\delta m \ll m_0$ and $\delta k \ll k_0$, so that the induced structural variation can be treated as a small perturbation to the system.

Under harmonic excitation, the frequency domain mechanical equation of motion is

$$\left[\mathbf{K}_0 + \delta\mathbf{K} + k_p \mathbf{b}_m \mathbf{b}_m^\top + i\omega \mathbf{C}_m - \omega^2 (\mathbf{M}_0 + \delta\mathbf{M})\right] \mathbf{u} = \theta \mathbf{b}_m V_p, \tag{17}$$

where $\mathbf{M}_0$ and $\mathbf{K}_0$ are the mass and stiffness matrices of the undamaged chain, respectively, $\mathbf{C}_m$ is the grounded damping matrix, and

$$\delta\mathbf{M} = \delta m\, \mathbf{e}_j \mathbf{e}_j^\top, \qquad \delta\mathbf{K} = \delta k\, \mathbf{b}_d \mathbf{b}_d^\top, \qquad \mathbf{b}_d = \mathbf{e}_{j+1} - \mathbf{e}_j. \tag{18}$$

With the discrete electromechanical host model established, we next examine how the structural damage is translated into perturbations of the NHT circuit. The piezoelectric transducer is connected to the two ends of the NHT circuit. In addition, a balancing impedance Z_b (see in Fig. 6) is connected in parallel with the piezoelectric patch to cancel the baseline contribution of the undamaged mechanical impedance and thereby guarantee the sensitive operating condition of the NHT circuit.

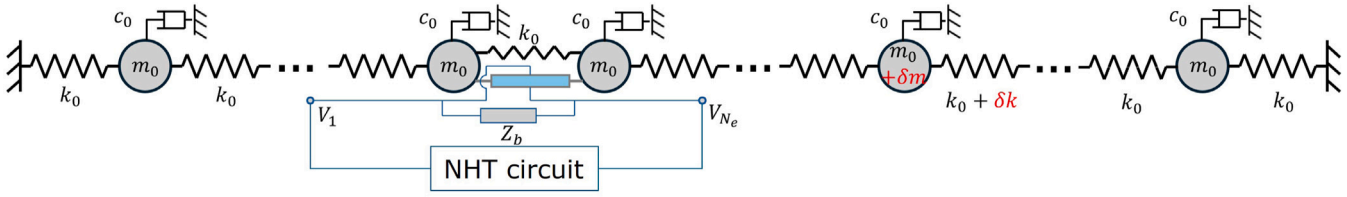


Fig. 6. The discrete model of the NHT-enhanced EMI method, where δm and δk represent the local mass and stiffness changes due to damage. The piezoelectric transducer is connected to boundary of the NHT circuit boundary.

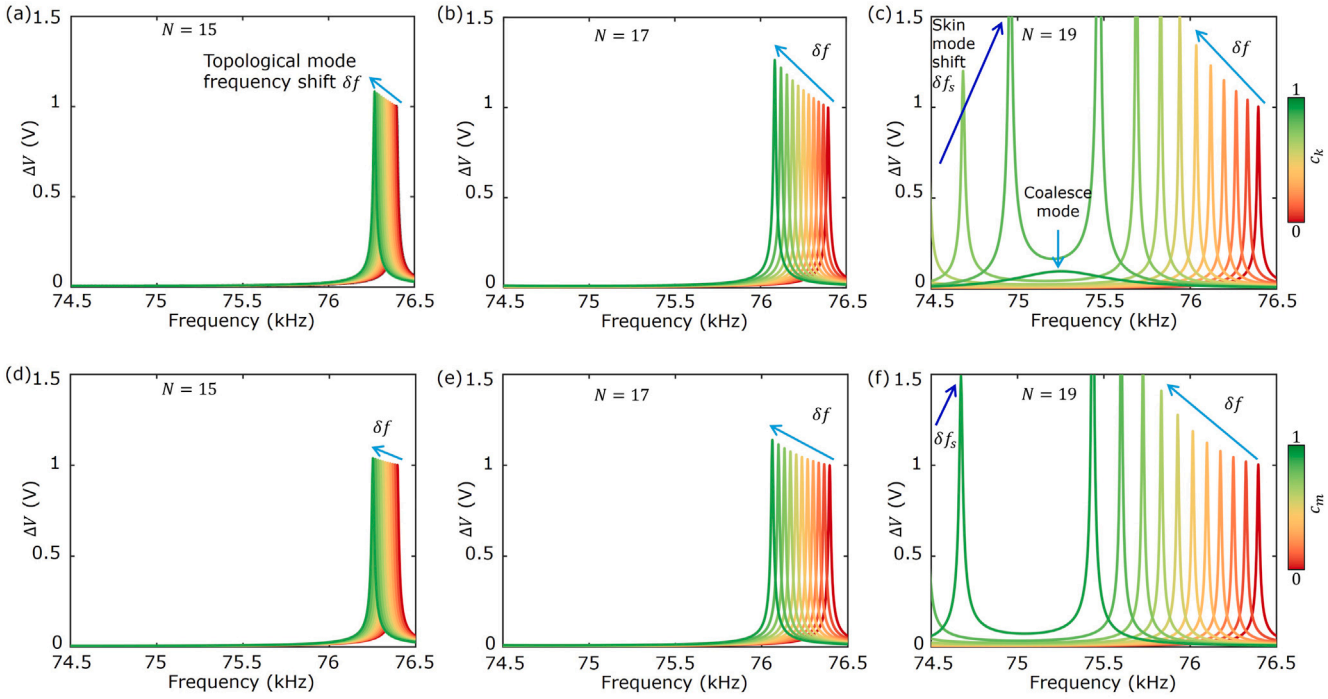


Fig. 7. Voltage response spectra of the NHT-enhanced EMI method. (a-c) Spectra of the voltage response δV for stiffness damages only with circuit sizes of (a) $N = 15$, (b) $N = 17$, and (c) $N = 19$. (d-f) Spectra of voltage response δV for mass damages only with circuit sizes of (d) $N = 15$, (e) $N = 17$, and (f) $N = 19$. c_k and c_m are normalized damage coefficients, serving as indicators of damage severity: $\delta k = c_k \delta k_0$, $\delta m = c_m \delta m_0$.

4.2. Perturbation characterization of topological frequency shift due to damages

The fully coupled electromechanical system is governed by the following equations of motion:

$$\begin{aligned} &\theta \mathbf{b}_m \mathbf{b}_e^\top \mathbf{V} = \mathbf{Z}_m(\omega)\mathbf{u}, \\ &\mathbf{P}(\omega)\mathbf{V} = \left[-\omega^2 C_p^T + i\omega Y_b(\omega)\right]\mathbf{B}\mathbf{V} - \omega^2 \theta \mathbf{b}_e \mathbf{b}_m^\top \mathbf{u}, \end{aligned} \tag{19}$$

where $\mathbf{P}(\omega) = -\omega^2\mathbf{C} + i\omega\mathbf{R} + \mathbf{L}$, and $Y_b(\omega) = 1/Z_b(\omega)$ is the balancing admittance.

Eliminating the mechanical degrees of freedom from Eq. (19) yields

$$\mathbf{P}(\omega)\mathbf{V} = \Pi(\omega)\mathbf{B}\mathbf{V}, \tag{20}$$

with an effective boundary impedance

$$\Pi(\omega) = -\omega^2 C_p^T + i\omega Y_b(\omega) - \omega^2\theta^2 z_m(\omega), \tag{21}$$

where $z_m(\omega) = \mathbf{b}_m^\top \mathbf{Z}_m^{-1}(\omega)\mathbf{b}_m$ is the projected mechanical compliance seen by the piezoelectric element.

We now use the first-order perturbation expansion about the NHT frequency $\omega_t = 2\pi f_t$, treating the damage-induced stiffness and mass matrix variations, $\delta\mathbf{K}$ and $\delta\mathbf{M}$, as small perturbation terms. Let

$$\omega = \omega_t + \delta\omega, \qquad \mathbf{V} = \mathbf{V}_R + \delta\mathbf{V}, \tag{22}$$

and expand the projected mechanical term as

$$z_m = z_{m,0} + \delta z_m + \mathcal{O}(\delta^2), \tag{23}$$

where the undamaged structural contribution is

$$z_{m,0} = \mathbf{b}_m^\top \mathbf{Z}_{m,0}^{-1}\mathbf{b}_m, \qquad \mathbf{Z}_{m,0} = \mathbf{K}_0 + k_p \mathbf{b}_m \mathbf{b}_m^\top - \omega_t^2 \mathbf{M}_0. \tag{24}$$

Using the standard first-order inverse expansion [78], we have

$$\delta z_m = \delta\hat{K} + \omega_t^2 \delta\hat{M} + 2\omega_t \delta\omega\, \hat{G}, \tag{25}$$

with $\delta\hat{K} = -\mathbf{b}_m^\top \mathbf{Z}_{m,0}^{-1}\delta\mathbf{K}\,\mathbf{Z}_{m,0}^{-1}\mathbf{b}_m$, $\delta\hat{M} = \mathbf{b}_m^\top \mathbf{Z}_{m,0}^{-1}\delta\mathbf{M}\,\mathbf{Z}_{m,0}^{-1}\mathbf{b}_m$, and $\hat{G} = \mathbf{b}_m^\top \mathbf{Z}_{m,0}^{-1}\mathbf{M}_0\mathbf{Z}_{m,0}^{-1}\mathbf{b}_m$.

Substituting these expansions into Eq. (20), the zeroth-order problem (δ^0) is

$$\left[\mathbf{P}(\omega_t) - \Pi_0(\omega_t)\mathbf{B}\right]\mathbf{V}_R = 0, \tag{26}$$

where $\Pi_0(\omega_t) = -\omega_t^2\left(C_p^T + \theta^2 z_{m,0}\right) + i\omega_t Y_b(\omega_t)$. Proceeding to the first order (δ^1), we obtain

$$\begin{aligned} &\left[\mathbf{P}(\omega_t) - \Pi_0(\omega_t)\mathbf{B}\right]\delta\mathbf{V} \\ &+ \left[-2\omega_t\delta\omega\mathbf{C} + i\delta\omega\mathbf{R} - \delta\Pi\,\mathbf{B}\right]\mathbf{V}_R = 0, \end{aligned} \tag{27}$$

where $\delta\Pi = [-2\omega_t(C_p^T + \theta^2 z_{m,0}) + iY_b(\omega_t) + i\omega_t Y_b'(\omega_t) - 2\omega_t^3\theta^2\hat{G}]\delta\omega - \omega_t^2\theta^2\left(\delta\hat{K} + \omega_t^2\delta\hat{M}\right)$. By multiplying both sides of the first-order equation with the left eigenvector $\mathbf{V}_L$ and considering the biorthogonal

condition, Eq. (27) can be written as

$$\mathbf{V}_L\left[(-2\omega_t\mathbf{C}+i\mathbf{R})\delta\omega-\delta\Pi\mathbf{B}\right]\mathbf{V}_R=0, \tag{28}$$

thus, the shifted topological frequency can be expressed as

$$\delta f=\frac{\omega_t^2\theta^2(\delta\hat{K}+\omega_t^2\delta\hat{M})\mathbf{V}_L\mathbf{B}\mathbf{V}_R}{\mathbf{V}_L\left[-2\omega_t\mathbf{C}+i\mathbf{R}-\delta\Pi\mathbf{B}\right]\mathbf{V}_R}. \tag{29}$$

Note that in the zeroth-order equation Eq. (26), the effective boundary impedance $\Pi_0(\omega_t)$ is of order δ^0, which means that it cannot be taken as a weak perturbation. Therefore, the response to any further damage-induced variation will be insensitive, thereby undermining the sensing design. To avoid this issue, $\Pi_0(\omega_t)$ must be canceled by properly tuning the balancing impedance Z_b (admittance $Y_b = 1/Z_b$). Accordingly, Y_b must satisfy

$$\begin{aligned}&\Pi_0(\omega_t)=0,\\&-2\omega(\mathbf{C}_p^T+\theta^2 z_{m,0})+iY_b+i\omega Y_b'-2\omega_t^3\theta^2\hat{G}=0.\end{aligned} \tag{30}$$

Without loss of generality, we assume an ideal balancing impedance in this subsection, while practical realization and local approximation of $Y_b(\omega)$ near ω_t can be found in Appendix H. Hence, the zeroth-order problem reduces to the unperturbed NHT eigenvalue problem,

$$\mathbf{P}(\omega_t)\mathbf{V}_R=0, \tag{31}$$

while the first-order correction is governed by the damage-induced terms $\delta\hat{K}$ and $\delta\hat{M}$. With ideal Y_b, Eq. (27) can be expressed as:

$$[(-2\omega_t\,\mathbf{C}+i\mathbf{R})\delta\omega+\omega_t^2\theta^2(\delta\hat{K}+\omega_t^2\delta\hat{M})\,\mathbf{B}]\mathbf{V}_R=0. \tag{32}$$

Harnessing the biorthogonality of the non-Hermitian system, projecting Eq. (32) onto the unperturbed left eigenvector $\mathbf{V}_L$ of the intact topological mode gives the first-order NHT frequency shift:

$$\delta f=\frac{\mathbf{V}_L^{\top}\left[\omega_t^2\theta^2\left(\delta\hat{K}+\omega_t^2\delta\hat{M}\right)\mathbf{B}\right]\mathbf{V}_R}{2\pi\,\mathbf{V}_L^{\top}\left[\partial\mathbf{P}(\omega)/\partial\omega\right]\mathbf{V}_R}. \tag{33}$$

Equation (33) has the same structure as Eq. (4), in which the NHT frequency shift is determined by the damage-induced perturbations $\delta\hat{K}$ and $\delta\hat{M}$, and also by the NHT circuit design.

4.3. Numerical validation of the NHT-enhanced EMI response

To measure the NHT frequency shift in the frequency domain, we apply harmonic excitation at the boundary node of the NHT circuit and evaluate the resulting voltage difference ΔV across the two electrodes of the piezoelectric transducer. In this case, the mechanical parameters are $k_0 = 10^9$ N/m, $c_0 = 1$ N·s/m, and $m_0 = 0.01$ kg. The parameters of the circuit we used in this model are $C_1 = 0.85$ nF, $C_2 = 0.35$ nF, and $C_3 = 1.65$ nF. All voltage nodes are grounded with a resistor $R_0 = 5$ MΩ, capacitor $C_0 = 2.5$ nF, inductor $L = 1.6$ mH. The piezoelectric patch is mounted between 3rd and 4th mass ($i = 3$), while the damage is added in 9th mass ($j = 9$). Figs. 7(a-c) show the ΔV in the vicinity of the topological frequency $f_t = 76.4$ kHz for stiffness damage only ($\delta M = 0$) with $N = 15$, 17, and 19, respectively. As the circuit size increases, the topological frequency shift δf is progressively amplified, demonstrating the size-enhanced sensitivity of the NHT circuit. Meanwhile, Fig. 7(c) shows that a skin mode shifts into the frequency range and coalesces with the topological mode. This observation confirms the predicted sensing limit: although increasing the number of unit cells enhances the boundary sensitivity, excessively strong sensitivity can also drive coalescence between the NHT mode and a nearby skin mode, thereby limiting the available sensing range. Figs. 7(d-f) show the ΔV in the vicinity of the topological frequency $f_t = 76.4$ kHz for mass damages only ($\delta K = 0$) with $N = 15$, 17, and 19, respectively. Similar to the stiffness case, δf is progressively amplified as N increases. In contrast to the stiffness damage results in Fig. 7(c), however, no evident coalescence with nearby skin modes is observed in the mass-perturbation case over the parameter range considered here, indicating that the topological mode remains isolated. To provide a fair comparison with conventional EMI, we include in Appendix C an RMSD-based assessment over the full spectrum for the conventional and proposed methods. These results further confirm that the proposed method provides a stronger damage-sensitive signature than the conventional baselines.

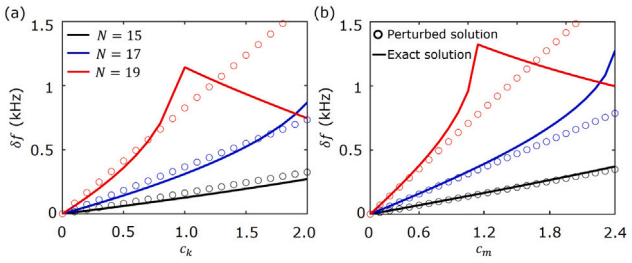


Fig. 8. Comparison of the topological frequency shift obtained using the discrete electromechanical NHT system and the first-order perturbation theory. (a) δf versus normalized stiffness perturbation c_k for $N = 15$, 17, and 19. Solid lines denote the exact solution of the coupled model, and open circles denote the perturbation-based prediction. (b) δf versus normalized mass perturbation c_m for $N = 15$, 17, and 19.

In addition, we use the exact solution to validate the first-order perturbation theory. The results can be found in Fig. 8(a,b) for stiffness and mass damages, respectively. The perturbation results show good agreement with the exact solution in the weak-perturbation regime. They also show progressive amplification of δf with increasing N. Note that the present formulation is derived within a linearized weak perturbation framework and therefore does not account for nonlinear material responses or hysteretic effects that may arise at larger damage levels or under severe environmental conditions. Extending the projected boundary perturbation to incorporate nonlinear constitutive behavior is an important direction for future work, but it lies beyond the scope of the current study.

5. NHT-enhanced damage detection in continuum

With the NHT-enhanced EMI method established in the previous sections, we now examine how this method performs in a continuous model. The purpose of this section is not to re-derive the circuit sensitivity law, but to validate that realistic structural damage can indeed be mapped onto measurable shifts in the topological frequency, and to compare this response with that of the conventional EMI method under the same host structure and damage conditions.

To numerically validate the practical sensing capability of the proposed NHT-enhanced EMI method, a one-dimensional Euler–Bernoulli cantilever beam with a bonded piezoelectric patch is investigated, as illustrated in Fig. 9. The beam has a length $L = 50$ mm, width $w = 10$ mm, and thickness $h = 1$ mm, while the piezoelectric patch mounted near the fixed end has a length $L_p = 2$ mm, width $w_p = 10$ mm, and height $h_p = 0.5$ mm. The top and bottom electrodes of the patch are connected across the ends of the NHT circuit. Fig. 9(b) shows the conventional EMI response in the intact and damaged cases for an artificial crack with $l_c = 0.5$ mm, $w_c = 0.1$ mm, and $h_c = 0.5$ mm. The material properties of the substrate and piezoelectric patch are provided in Appendix D. A frequency shift of approximately 150 Hz is observed in the EMI spectrum when $\Delta x = 15$ mm, which serves as a benchmark for comparison.

To further examine the response under different damage scenarios, Figs. 9(c–e) compare the conventional and NHT-enhanced EMI results for cracks located at $\Delta x = 15$, 30, and 45 mm, respectively. We use an ideal Z_b to cancel the inherent capacitance value and undamaged mechanical impedance. In the proposed NHT-enhanced EMI method, the primary sensing observable is the topological mode frequency shift,

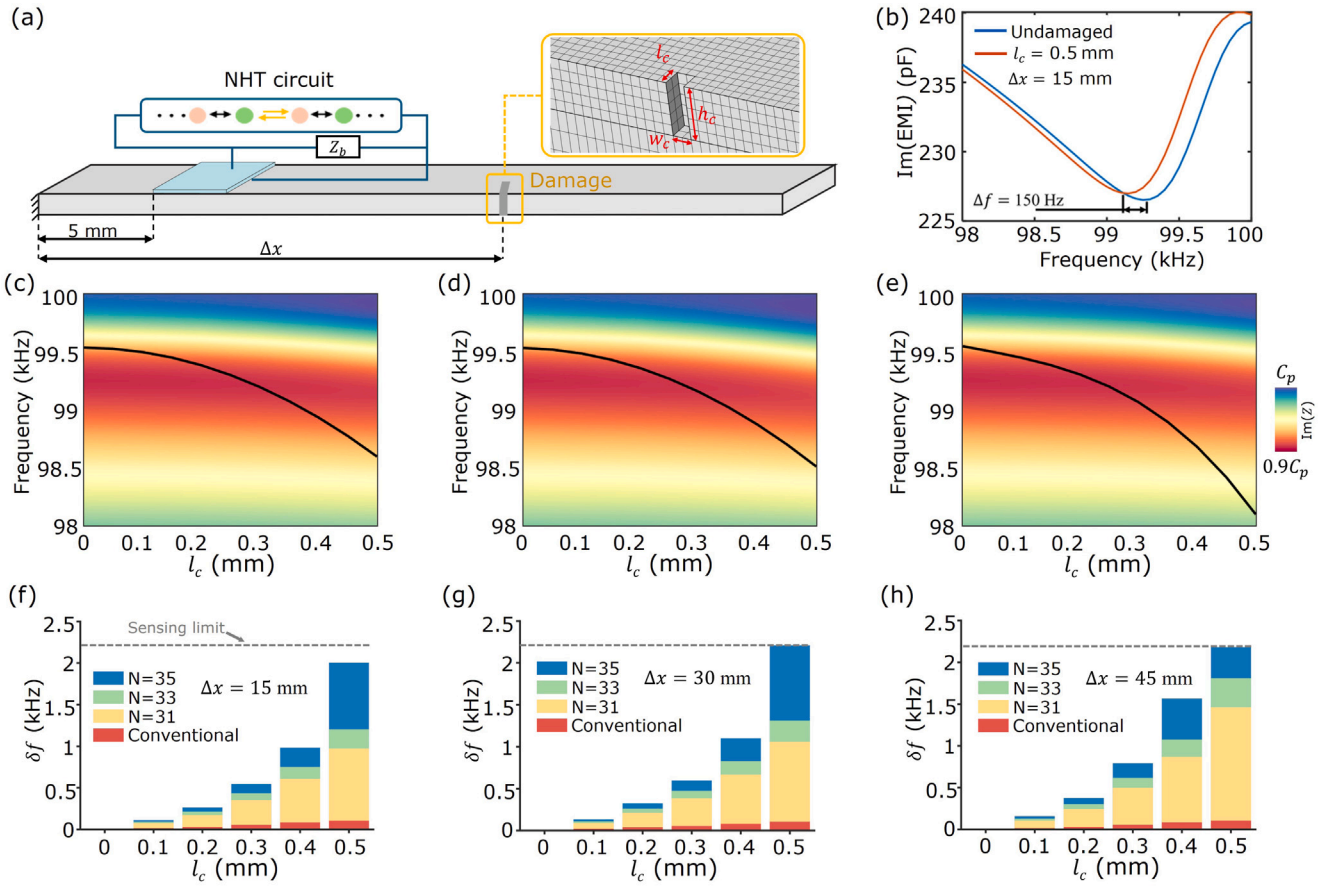


Fig. 9. NHT-enhanced EMI method for crack detection on a cantilever beam. (a) Configuration of the cantilever beam and the piezoelectric patch. (b) Frequency shift of the conventional EMI method. (c-e) The conventional EMI spectra (contour plots) and NHT frequencies (black curves) under different damage lengths l_c when the damage is located at (c) Δx = 15 mm, (d) Δx = 30 mm, and (e) Δx = 45 mm. (f-h) NHT frequency shifts with different circuit sizes and comparison with the conventional EMI method, when (f) Δx = 15 mm, (g) Δx = 30 mm, and (h) Δx = 45 mm.

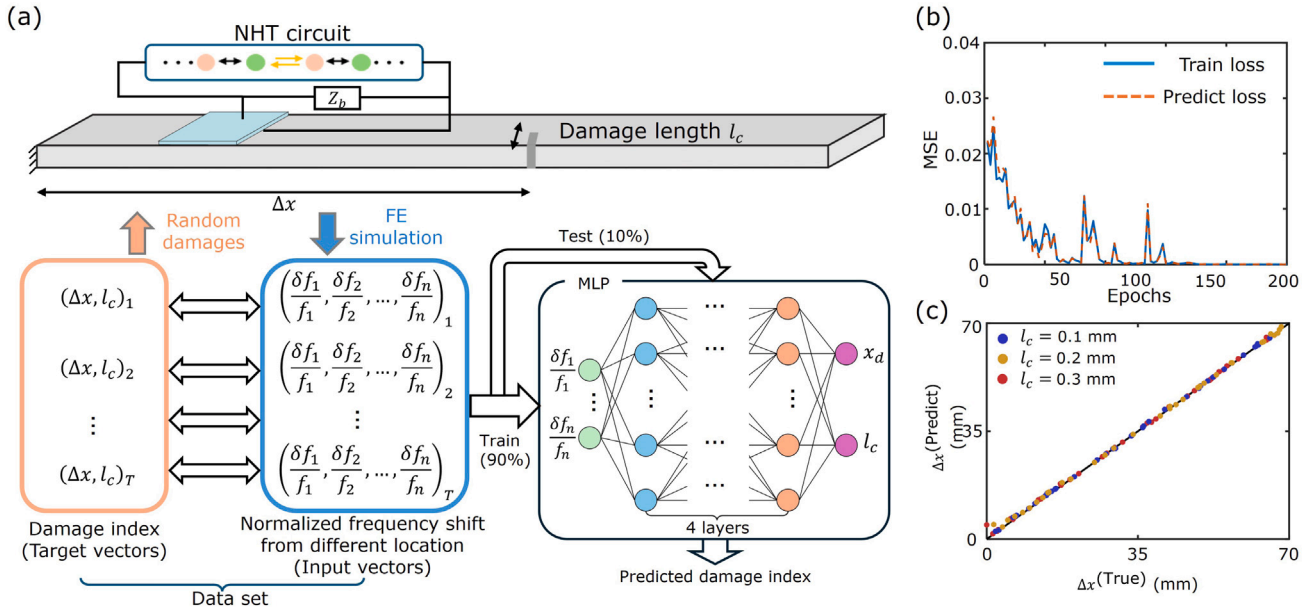


Fig. 10. The framework of the NHT enhanced EMI method for damage localization and severity evaluation in a 1D cantilever beam by an MLP. (a) Configuration of the NHT-enhanced EMI method on cantilever beam. The target vector includes the location and severity, and the input vector includes the frequency shifts under different topological frequencies. In this case, we set $n = 9$, and the topological frequencies are chosen as $[f_1, \ldots, f_9]$=[85, 86, 87, 88, 89, 90, 91, 92, 93] kHz. (b) The training loss and prediction loss in training process. (c) The predicting results with different damage severities.

shown by the black lines, while the background colormaps represent the corresponding variation of the effective capacitance spectrum. Under the same structural perturbation, the NHT-enhanced configuration exhibits substantially larger topological frequency shifts. For example, at $\Delta x = 15$ mm, a shift of nearly 950 Hz is obtained using a circuit with $N = 31$. When the crack location moves to $\Delta x = 30$ mm and $\Delta x = 45$ mm, the corresponding topological frequency shifts increase to about 1050 Hz and 1450 Hz, respectively. These results indicate a much stronger damage-sensitive frequency response in the NHT-enhanced method in terms of frequency-shift magnitude. Specifically, in the most favorable case considered here, the topological frequency shift is approximately 10 times larger than the frequency shift of the conventional EMI method. We note, however, that this comparison is based on the magnitude of a selected frequency shift observable, rather than on a detection limit defined under noise, drift, or estimator uncertainty. In the most favorable case considered here, the topological frequency shift is approximately 10 times larger than that of the selected conventional EMI peak. Figs. 9(f–h) further show that increasing the circuit size N enhances the damage-induced topological frequency shift. However, when $\Delta x = 45$ mm and $l_c = 0.5$ mm, the amplification is not exponential but rather linear. In this case, the topological frequency approaches a nearby skin mode and undergoes coalescence, which makes it less sensitive to circuit size.

6. MLP-assisted damage localization

Although the cantilever beam has demonstrated the sensing capability of the NHT-enhanced EMI method, a single topological frequency shift is generally insufficient to characterize both the location and severity of the structural damage. As shown by Eq. (25), the projected electromechanical perturbation depends not only on the damage-induced variations $\delta\mathbf{K}$ and $\delta\mathbf{M}$, but also on how these variations are filtered by the intact structural resolvent $\mathbf{Z}_{m,0}^{-1}(\omega)$ and by the boundary coupling vector $\mathbf{b}_m$. Consequently, the measured topological frequency shift is intrinsically frequency dependent and location dependent: different operating frequencies probe different dynamic projections of the same damage, while different sensing locations produce different boundary-sensitive responses to the same damage. This motivates the use of a multi-feature sensing strategy, in which topological frequency shifts collected at several designed operating topological frequencies and/or sensing locations are assembled into an enriched feature vector. Here, we employ multiple NHT sensing channels with different designed topological frequencies and use their combined responses as the input to a multilayer perceptron (MLP). In this way, the MLP is introduced as an inverse model that maps a set of physics-based NHT observables to the corresponding damage indicators.

According to Eq. (2), the topological frequency $f_t = (2\pi\sqrt{LC_0})^{-1}$ can be tuned through the circuit parameters, such as the inductance L [73]. By selecting multiple operating frequencies, one obtains different sensitivity distributions over the host structure. The resulting multi-frequency response provides a richer description of the structural state than a single NHT frequency shift, which enables the inverse identification of damage.

For the cantilever beam case, the input vector is defined as $\mathbf{x} = \left[\frac{\Delta f_1}{f_1}, \frac{\Delta f_2}{f_2}, \ldots, \frac{\Delta f_n}{f_n}\right]^\top$, where $\Delta f_i/f_i$ denotes the normalized shift of the ith topological operating frequency. The output target vector is chosen as $\mathbf{y} = [\Delta x,\ l_c]^\top$, where Δx is the crack location and l_c is the crack length used to characterize the damage severity. In the present study, $n = 9$, and the nine operating frequencies are selected in the range of 85 to 93 kHz. The feature vector is extracted from finite-element simulations of randomly generated damage cases. The crack width and depth are fixed, while the crack location is varied along the beam, and the crack length takes three discrete values. A total of 8000 samples are generated, among which 90% are used for training, and the remaining 10% are reserved for testing and validation. The dataset is normalized by z-score standardization. In addition, a validation subset is separated from the training data to monitor the optimization process and reduce overfitting.

The MLP architecture for the beam case has 6 hidden layers, and each layer has 256 neurons with GELU activation [79–82]. The input neurons correspond to the n normalized topological frequency shifts, and the two output neurons represent the target damage parameters ($[\Delta x,\ l_c]^\top$). The model is trained by minimizing the mean-squared error (MSE) using the Adam optimizer with a learning rate 10^{-3}, a batch size of 32, and 200 training epochs, while the validation subset is used to track the convergence behavior and to reduce the risk of overfitting. A step learning rate scheduler is applied every 80 epochs with a rate of 0.2. The equation of MSE is

$$\mathrm{MSE} = \frac{1}{N_s}\sum_{k=1}^{N_s}\left\|\mathbf{y}_k^{\mathrm{pred}} - \mathbf{y}_k^{\mathrm{true}}\right\|_2^2, \tag{34}$$

where $\mathbf{y}_k^{\mathrm{pred}}$ and $\mathbf{y}_k^{\mathrm{true}}$ denote the predicted and reference target vectors of the kth sample, respectively, and N_s is the number of samples. Fig. 10(a) illustrates the overall framework, and Fig. 10(b) shows the corresponding training history. Both the training and validation losses decrease rapidly and then approach stable values, indicating that the inverse mapping can be learned effectively from the present dataset.

The beam predictions are shown in Fig. 10(c). The predicted crack locations and severity agree closely with the corresponding finite-element reference values, and the resulting R^2 values are approximately equal to 1.00. These results indicate that, under the present numerical conditions, the multi-channel NHT frequency shifts contain sufficient information for the simultaneous identification of damage position and severity. At the same time, it should be emphasized that the current dataset is generated from controlled FE simulations without measurement noise. Therefore, the near-unity prediction accuracy should be interpreted as a proof-of-concept demonstration of identifiability under the modeled conditions, rather than as a claim of universal robustness in practical deployment.

The same framework can be extended naturally to higher-dimensional structures by employing multiple distributed piezoelectric transducers for multiple NHT sensing channels. As indicated by Eq. (25), the EMI variation is governed not only by frequency, but also by the spatial location of the piezoelectric transducer through the projected mechanical response of the host structure. In the 1D beam example above, we primarily exploit the frequency dependence of the NHT-enhanced EMI response to construct the input features. In the 2D plate case considered here, we further incorporate the spatial dependence by using multiple piezoelectric patches distributed over the structure. The detailed locations of the 12 piezoelectric patches on the plate are listed in Table 3. As illustrated in Fig. 11(a), the normalized topological frequency shifts collected from different sensing locations are combined into an input vector, while an artificial rectangular microperforation is randomly introduced in the plate model to represent structural damage. The MLP is then trained to predict the corresponding damage coordinates (x_d, y_d). Similar to the beam case, the input features are extracted from finite-element simulations, while the output target is defined as $\mathbf{y} = [x_d,\ y_d]^\top$. A total of 6400 different damage locations are randomly generated on the plate, and the shifted topological frequencies are calculated from the eigenvalue analysis. The training and testing datasets are split similarly to those in the 1D setting. The hyperparameters and the architecture of the MLP hidden layers remain unchanged.

Fig. 11(b) shows the training process for the plate model. The training and validation losses again converge after approximately 200 epochs, indicating stable learning of the inverse relation between the NHT frequency shift vector and the damage coordinates. To quantify the localization error, we calculate the Euclidean distance between the predicted and true damage positions,

$$\epsilon = \sqrt{\left(x_d^{\mathrm{pred}} - x_d^{\mathrm{true}}\right)^2 + \left(y_d^{\mathrm{pred}} - y_d^{\mathrm{true}}\right)^2}. \tag{35}$$

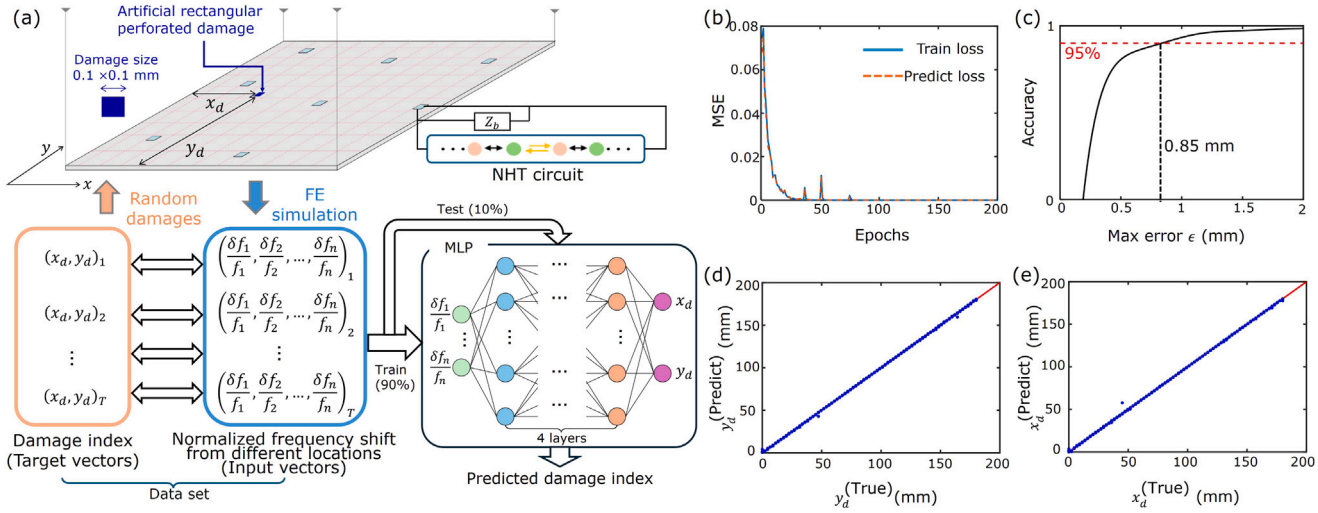


Fig. 11. The framework of the NHT-enhanced EMI method for damage localization and severity evaluation in a 2D host structure by an MLP. (a) Configuration of NHT-enhanced EMI method on a plate. Target vector includes the coordinates of location, and input vector includes the frequency shifts at different locations. In this case, we set $n = 12$, and the unperturbed topological frequency is chosen as 95 kHz. (b) The training loss and prediction loss during the training process. (c) The accuracy of the predicted results. The MLP can predict 95% of the cases with a maximum error of 0.85 mm. (d) The predicting results of the x coordinate. (d) The predicting results of the y coordinate.

The localization results are summarized in Fig. 11(c-e). In the present study, the trained MLP achieves a 95% prediction accuracy for $\epsilon < 0.85$ mm. Since the plate size is 180×180 mm^2, this error corresponds to less than 0.5% of the plate length, indicating submillimeter localization capability relative to the structural dimension and the ability to detect and distinguish localized damage.

The present MLP framework should be interpreted within the scope of the dataset used for training. In this work, the training samples are generated under a fixed structural model, fixed sensor arrangement, and fixed boundary conditions. As a result, the learned inverse mapping is conditionally valid for the class of damage scenarios represented in the training data. If the boundary conditions change, or if the damage morphology departs significantly from the patterns included in the dataset, the measured frequency shift vector may move outside the learned distribution and the prediction accuracy can deteriorate. Likewise, the localization capability depends on the richness of the sensing inputs, which is governed by the number and distribution of sensing channels. A denser sensing layout generally provides a more distinctive multi-frequency signature and therefore improves inverse identification accuracy. Additional analyses of dataset statistics, feature contribution, training-size dependence, and noise sensitivity are provided in Appendices E and F.

7. Discussion and conclusion

This work proposes an NHT-enhanced electromechanical impedance framework for damage sensing, in which the piezoelectric EMI response of the host structure is recast as an effective boundary perturbation of an NHT circuit. Under this formulation, structural damage is converted into a shift of the topological frequency, thereby providing a new sensing observable beyond the conventional EMI spectrum-based interpretation. This work establishes the theoretical and numerical framework that links structural damage, the electromechanical impedance method, and the non-Hermitian topological sensor. The discrete electromechanical model, together with the beam and plate examples, provides conceptual validation of this mechanism across different levels of modeling complexity. The results show that, under the same structural perturbation, the proposed NHT architecture can produce a significant frequency shift response compared to the conventional EMI method.

Specifically, we numerically probe the sensing limitations of the proposed method. The usable sensing range is restricted by the possible interaction or coalescence between the tracked topological mode and nearby skin modes. In general, increasing the designed sensitivity causes the topological mode to reach this sensing limit under a smaller perturbation, revealing an inherent trade-off between sensitivity enhancement and the available sensing range. It should be emphasized that the present study is based on idealized circuit elements and numerically generated structural responses, and therefore does not yet fully capture experimental non-idealities such as parasitic effects, drift, component tolerances, active-element limitations, or environmental variability. The supplementary noise analyses in Appendix F are included to provide controlled numerical evidence that the proposed sensing mechanism is not immediately destroyed by moderate perturbations; however, these results remain simplified tests and do not constitute a realistic hardware-level validation. The machine-learning results should likewise be understood within this proof-of-concept scope: they confirm that the topological frequency shift features contain sufficient damage information in the present numerical setting, but further experimental validation is still required to assess robustness under different types of material, noise, boundary-condition uncertainty, and distribution shifts.

The proposed framework is not restricted to a particular host geometry or an idealized defect model, but can be applied to more general structural configurations and damage cases, provided that the associated mechanical variation can be projected into the piezo-mediated electromechanical boundary response. At the same time, identifying the specific damage type generally requires a richer dataset and additional discriminatory features, since different physical damage mechanisms may produce similar effective perturbations and therefore similar topological frequency shifts. Therefore, the value of the present work lies in establishing a general sensing principle: non-Hermitian topological circuits can serve as an amplification layer for electromechanical damage indicators, converting weak structural perturbations into measurable topological frequency shifts. This provides a foundation for future studies on experimental implementation and robust damage identification in practical structural health monitoring systems.

CRediT authorship contribution statement

Honghua Qian: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Jiaji Chen:** Writing – review & editing, Writing – original

draft, Supervision, Software. **Shaoyun Wang:** Methodology. **Nan Shao:** Software. **Qian Wu:** Methodology. **Guoliang Huang:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Details of derivation

This appendix provides the explicit circuit matrices used in the theoretical formulation and the detailed derivation of the asymptotic sensitivity expression employed in Eq. (8).

A.1. Circuit matrices and state vector

Some of the matrices and vectors appearing in the theoretical section of the manuscript are given explicitly below. In Eq. (1), the inductance matrix is

$$0.8\mathbf{L} = \begin{pmatrix} 1/L & 0 & \dots & 0 & 0 \\ 0 & 1/L & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & & 1/L & 0 \\ 0 & 0 & \dots & 0 & 1/L \end{pmatrix}_{N_e \times N_e}, \tag{36}$$

and the capacitance matrix is

$$0.8\mathbf{C} = \begin{pmatrix} C_0 & -C_2 & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\ -C_3 & C_0 & -C_1 & \ddots & & & & \vdots \\ 0 & -C_1 & C_0 & -C_2 & \ddots & & & \vdots \\ \vdots & \ddots & -C_3 & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & -C_1 & \ddots & \vdots \\ \vdots & & & \ddots & -C_1 & C_0 & -C_2 & 0 \\ \vdots & & & & \ddots & -C_3 & C_0 & -C_1 \\ 0 & \cdots & \cdots & \cdots & \cdots & 0 & -C_1 & C_0 \end{pmatrix}_{N_e \times N_e}. \tag{37}$$

In the ideal unperturbed circuit, the parasitic resistance is neglected, so that $\mathbf{R} = \mathbf{0}$.

For the odd-size chain considered here, $N_e = 2N-1$, and the voltage vector is written as

$$\mathbf{V} = \left[V_A^{(1)}, V_B^{(1)}, \dots, V_A^{(N-1)}, V_B^{(N-1)}, V_A^{(N)}\right]^{\top}. \tag{38}$$

A.2. Asymptotic derivation of the sensitivity law

In Eq. (8), an asymptotic expression is used for the frequency shift under the assumption $N \gg 1$. The derivation is given below.

We start from the exact first-order frequency shift of the topological mode under a boundary capacitance perturbation:

$$\delta\omega_t = 4\pi^3 f_t^3 L\, C_p\, a_l a_r\, \mathcal{B}, \tag{39}$$

where $s = (N-1)/2$, and the boundary interference factor is

$$\mathcal{B} = 1 + (\lambda_l \lambda_r)^s - \lambda_r^s - \lambda_l^s = (1-\lambda_l^s)(1-\lambda_r^s). \tag{40}$$

For the right-localized skin-effect regime, one has $|\lambda_r| > 1$, $|\lambda_l| < 1$. For sufficiently large N, since $|\lambda_l|^s \ll 1$, one obtains

$$1 - \lambda_l^s \approx 1. \tag{41}$$

Meanwhile, because $|\lambda_r|^s \gg 1$, $1 - \lambda_r^s \approx -\lambda_r^s$. Therefore, $\mathcal{B} \approx -\lambda_r^s$, and the frequency shift becomes

$$\begin{aligned} \delta\omega_t &\approx 4\pi^3 f_t^3 L\, C_p\, a_l a_r\, (-\lambda_r^s) \\ &= 4\pi^3 f_t^3 L\, C_p\, a_l a_r\, (-\lambda_r^{-1/2}) \exp\left(\frac{\ln \lambda_r}{2} N\right). \end{aligned} \tag{42}$$

Similarly, for the left-localized skin-effect regime, $|\lambda_l| > 1$, $|\lambda_r| < 1$, so that $\mathcal{B} \approx -\lambda_l^s$, which yields

$$\delta\omega_t \approx 4\pi^3 f_t^3 L\, C_p\, a_l a_r\, (-\lambda_l^{-1/2}) \exp\left(\frac{\ln \lambda_l}{2} N\right). \tag{43}$$

These two cases can be summarized in the form

$$\delta\omega_t \approx C_p(\kappa_t \beta) e^{\alpha N}, \tag{44}$$

where $\kappa_t = 4\pi^3 f_t^3 L\, a_l a_r$, and

$$[\alpha, \beta] = \begin{cases} \left[\dfrac{\ln \lambda_l}{2}, -\lambda_l^{-1/2}\right], & |\lambda_r| < 1 < |\lambda_l|, \\ \left[\dfrac{\ln \lambda_r}{2}, -\lambda_r^{-1/2}\right], & |\lambda_l| < 1 < |\lambda_r|. \end{cases} \tag{45}$$

The approximation error can also be quantified explicitly. For the right-localized case, taking the dominant term as $\mathcal{B}_{\text{dom}} = -\lambda_r^s$, the residual is

$$\Delta\mathcal{B} = \mathcal{B} - \mathcal{B}_{\text{dom}} = 1 + (\lambda_l \lambda_r)^s - \lambda_l^s. \tag{46}$$

Hence, the relative error satisfies

$$\frac{|\Delta\mathcal{B}|}{|\mathcal{B}_{\text{dom}}|} \le |\lambda_r|^{-s} + |\lambda_l|^s + \left|\frac{\lambda_l}{\lambda_r}\right|^s. \tag{47}$$

Because $|\lambda_r| > 1$ and $|\lambda_l| < 1$, all three terms decay exponentially with N. Therefore, the asymptotic form becomes increasingly accurate as the system size increases and the mode localization becomes stronger. The same argument applies to the left-localized case by exchanging l and r.

Meanwhile, consider the leading order perturbation of frequency term

$$\frac{1}{\omega_t^2 L} - \frac{1}{(\omega_t + \delta\omega)^2 L} \approx -C_p a_l a_r \lambda_r^{\frac{-1}{2}} e^{\alpha N}, \tag{48}$$

the left side of upon equation can be written as

$$\frac{1}{\omega_t^2 L} - \frac{1}{(\omega_t + \delta\omega)^2 L} \approx \frac{-2}{\omega_t^3 L} \delta\omega. \tag{49}$$

The desired frequency shift of the topological mode can be written as

$$\delta\omega \approx C_p \frac{-\omega_t^3 L}{2} a_l a_r \lambda_r^{\frac{-1}{2}} e^{\ln(\lambda_r) N/2}, \tag{50}$$

when $|\lambda_r| < 1 < |\lambda_l|$, λ_r should be replaced by $|\lambda_l|$.

Appendix B. Mode coalescence and sensing limit

As shown in Fig. 12, the sensing limit of the proposed NHT-enhanced EMI method originates from the coalescence between the topological mode and its neighboring skin mode, based on the NHT circuit in Section 3. For voltage nodes $N_e = 31$, the topological mode frequency varies approximately linearly with the boundary perturbation in the small-perturbation regime (sensing range), which enables a stable and high-resolution sensing response. However, as the perturbation strength further increases, the topological branch approaches the nearest skin mode branch and eventually coalesces with it. After this point, the resulting hybridized coupled mode becomes weakly sensitive to further perturbation, leading to the saturation of the measurable frequency shift and thus defining the sensing limit (perturbation range is $C_p \in [-0.35 \quad 0.35]$ pF). When the circuit size is increased to $N_e = 37$,

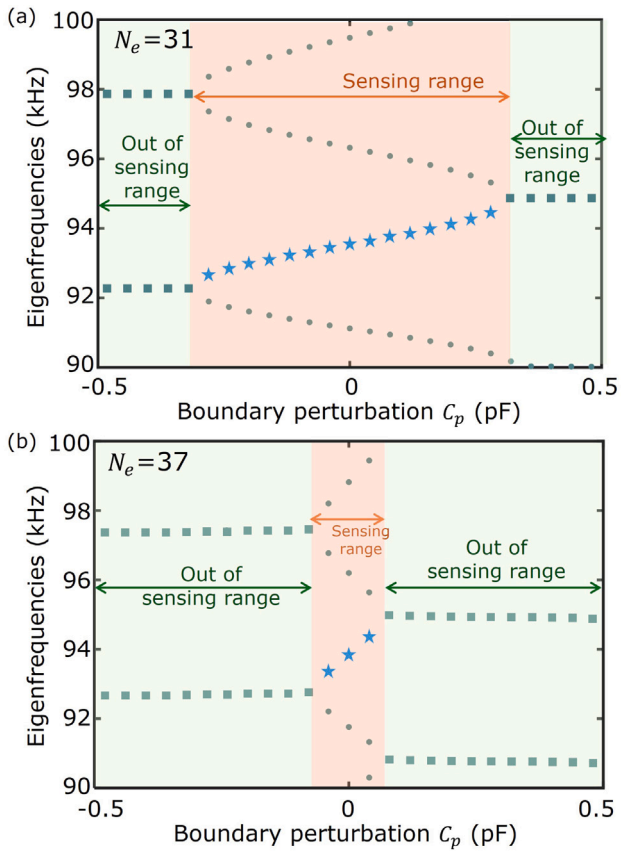


Fig. 12. Mode coalescence process under perturbation and the formation of sensing limits. (a) Eigenfrequency evolution and sensing range for the circuit with $N_e = 31$. (b) Corresponding results for $N_e = 37$. The mechanism of sensing limits. The blue pentagrams represent the topological mode, the green circles represent the skin modes, and the green squares represent the coalesced modes.

the same coalescence mechanism remains, but it occurs at a smaller perturbation level (perturbation range is $C_p \in [-0.04 \quad 0.04]$ pF). This is because increasing the circuit size amplifies the perturbation-induced topological frequency shift, thereby enhancing sensitivity, while at the same time driving the topological mode more rapidly toward the adjacent skin mode. As a result, although a larger N_e yields a stronger frequency shift response, it also narrows the available sensing range before mode coalescence occurs. Therefore, the circuit size must be selected by balancing these two competing factors, namely, sensitivity enhancement and sensing-range reduction. Therefore, the circuit size should be selected by balancing sensitivity enhancement against sensing-range reduction.

Appendix C. Spectrum comparison of proposed NHT-enhanced EMI method with two conventional EMI methods

In our proposed NHT-enhanced EMI method, we use the single resonance shift of electrical topological mode as the index reflecting damage, while the conventional EMI methods focus on the whole EMI spectrum variation. To address this consideration, we consider two more EMI configurations has same mechanical part with the proposed discrete model in Fig. 6, as shown in Fig. 13. The first model is the conventional EMI configuration shown in Fig. 6(a), in which a resistor R_0 is connected in series with the piezoelectric element and the branch is driven by a harmonic voltage source V_e [83]. Under harmonic excitation, the branch current is $I = i\omega Q_p$, and Kirchhoff's voltage law gives

$$V_e - V_p = i\omega R_0 Q_p. \tag{51}$$

By combining Eq. (17) with Eq. (51) and eliminating the mechanical degrees of freedom, the effective electrical impedance of the conventional EMI branch is obtained as

$$Z_{\mathrm{EMI}}(\omega) = R_0 + \left[i\omega \left(C_p^T - \theta^2 \mathbf{b}_m^\top \mathbf{Z}_m^{-1}(\omega) \mathbf{b}_m \right) \right]^{-1}, \tag{52}$$

where

$$\mathbf{Z}_m(\omega) = \mathbf{K}_0 + \delta\mathbf{K} + k_p \mathbf{b}_m \mathbf{b}_m^\top + i\omega\mathbf{C} - \omega^2(\mathbf{M}_0 + \delta\mathbf{M}) \tag{53}$$

is the mechanical dynamic-stiffness matrix seen by the piezoelectric element.

Equation (52) shows that the stiffness and mass perturbations, $\delta\mathbf{K}$ and $\delta\mathbf{M}$, enter the measured impedance through the projected mechanical term

$$z_m(\omega) = \mathbf{b}_m^\top \mathbf{Z}_m^{-1}(\omega) \mathbf{b}_m. \tag{54}$$

Accordingly, structural damage appears as a perturbation of the conventional EMI spectrum, primarily through changes in the reactive component near mechanical resonances. In the following, we therefore use $\mathrm{Im}(Z_{\mathrm{EMI}})$ as the conventional EMI quantity for comparison.

Fig. 14 summarizes the conventional EMI response of the discrete electromechanical model under localized stiffness and mass perturbations. Fig. 14(a) shows the full-spectrum $\mathrm{Im}(Z_{\mathrm{EMI}})$ for the intact, stiffness-damaged, and mass-damaged cases, and Fig. 14(b) provides enlarged views near the 2nd, 4th, 6th, and 10th resonances. Here, we focus on the imaginary part of the EMI impedance because it primarily reflects the reactive electromechanical interaction between the piezoelectric transducer and the host structure, and therefore provides a clearer representation of damage-induced resonance variation. In addition, $\mathrm{Im}(Z_{\mathrm{EMI}})$ is directly related to the effective capacitance of the piezoelectric subsystem, making it a standard and physically interpretable quantity for EMI-based damage evaluation. As shown in Fig. 14(a,b), the damage-induced variation is strongly mode dependent, with different resonances exhibiting different levels of peak shift and lineshape change.

To further compare the proposed scheme with a representative enhancement strategy in conventional EMI, we next consider the negative-capacitance (NC)-enhanced configuration shown in Fig. 13(b), where an NC element is connected in parallel with the piezoelectric patch [84,85]. In this case, the same discrete electromechanical host structure is retained, and only the external electrical compensation is modified. The NC branch effectively reduces the net capacitance seen by the piezoelectric element, thereby amplifying the damage-induced spectral variation in the EMI response. Since the NC branch shares the same terminal voltage V_p, the electrical balance is modified from Eq. (51) to

$$V_e - V_p = i\omega R_0 \left(Q_p + Q_{\mathrm{NC}} \right), \tag{55}$$

with $Q_{\mathrm{NC}} = C_N V_p$. Eliminating the mechanical degrees of freedom in the same manner yields the NC-enhanced branch impedance

$$Z_{\mathrm{EMI}}^{\mathrm{NC}} = R_0 + \left[i\omega \left(C_p^T + C_N - \theta^2 \mathbf{b}_m^\top \mathbf{Z}_m^{-1}(\omega) \mathbf{b}_m \right) \right]^{-1}. \tag{56}$$

From Eq. (56), the NC branch modifies the effective electrical capacitance seen by the piezoelectric element from C_p^T to $C_p^T + C_N$. Since the intrinsic piezoelectric capacitance contributes a background reactive term that can mask the projected mechanical variation $\theta^2 \mathbf{b}_m^\top \mathbf{Z}_m^{-1}(\omega) \mathbf{b}_m$, partial cancellation by the NC branch reduces this background contribution and makes the damage-induced impedance variation more pronounced. Therefore, the role of NC compensation is to amplify the conventional EMI spectral signature through capacitance balancing. However, the enhanced response remains spectrum-based and distributed over multiple resonances, rather than being converted into a single localized indicator.

Figs. 14(c)–(f) further quantify the spectrum variation using the root-mean-square deviation (RMSD) evaluated over a local frequency

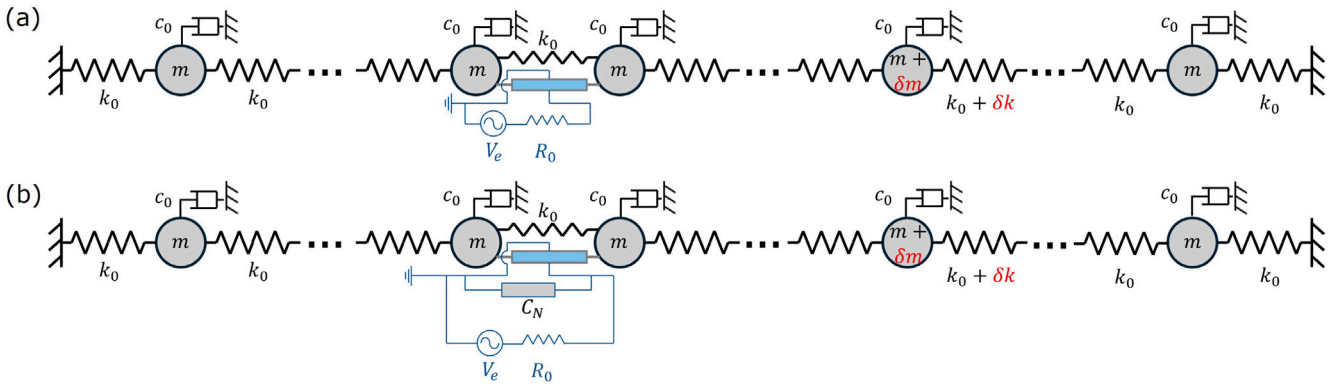


Fig. 13. Discrete models of two conventional EMI methods. (a) Conventional EMI method, where damage is inferred directly from broadband impedance spectrum variation. (b) Negative-capacitance-enhanced (NC-enhanced) EMI method. The value of the negative capacitor is set as $C_N = -0.98C_p^{\mathrm{T}}$.

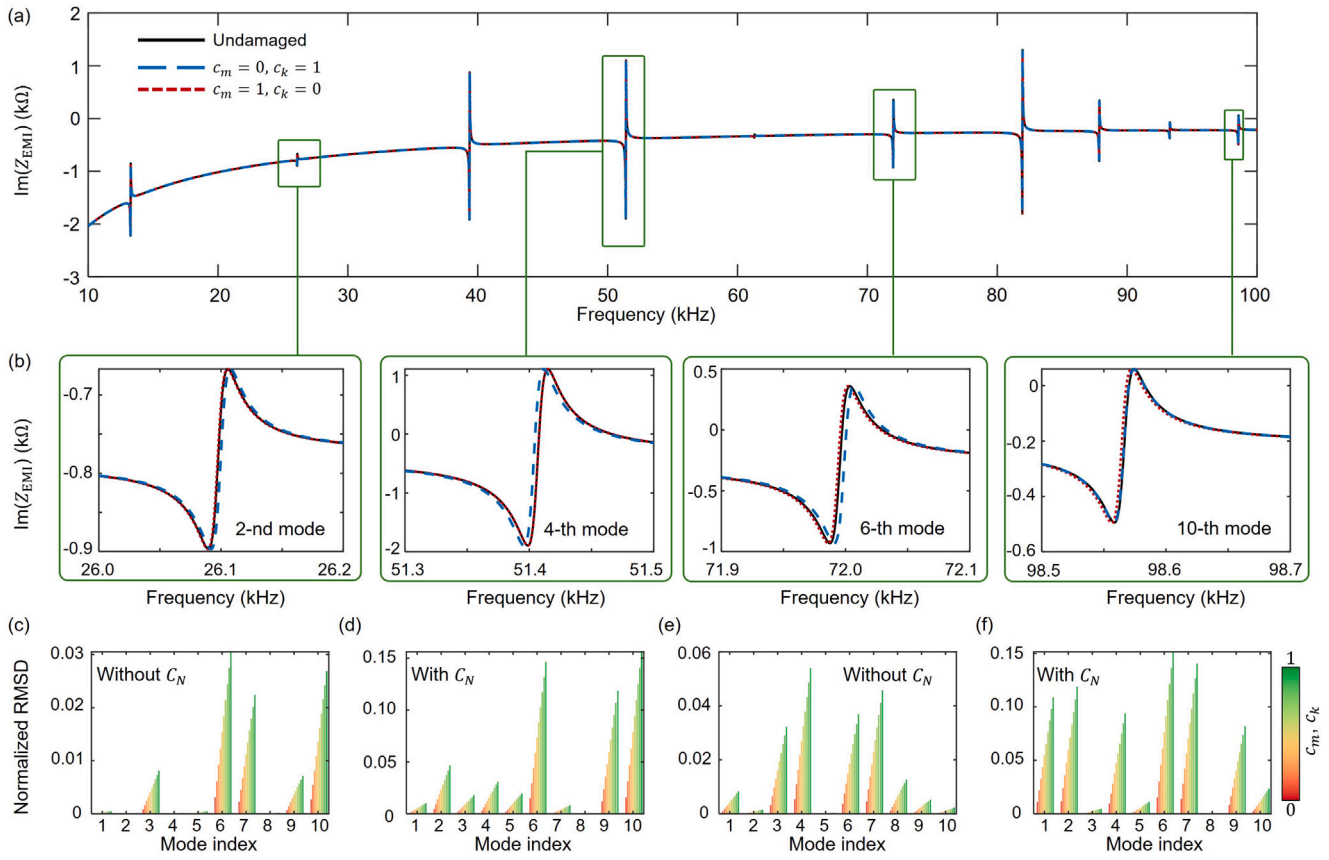


Fig. 14. Conventional EMI response of the discrete electromechanical model under stiffness and mass damage. (a) Full-spectrum Im(Z_{EMI}) for the intact structure, the stiffness-damage case ($c_m = 0, c_k = 1$), and the mass-damage case ($c_m = 1, c_k = 0$). The highlighted windows indicate the resonances selected for detailed comparison. (b) Enlarged views of the 2nd, 4th, 6th, and 10th EMI resonances. (c, d) Normalized RMSD of the conventional EMI spectrum variation under stiffness perturbation, without and with negative-capacitance compensation, respectively. (e, f) Normalized RMSD under mass perturbation, without and with negative-capacitance compensation, respectively. The color bar denotes the normalized damage coefficient c_m in (c, d) and c_k in (e, f).

window around each EMI resonance. Consistent with the spectral quantity adopted in this section, the RMSD is defined on $S(f) = \mathrm{Im}(Z_{\mathrm{EMI}}(f))$ as

$$\mathrm{RMSD} = \sqrt{\sum_{n=1}^{N_f} \left[S_d(f_n) - S_0(f_n)\right]^2 / \sum_{n=1}^{N_f} S_0(f_n)^2}, \tag{57}$$

where $S_0(f_n)$ and $S_d(f_n)$ denote the intact and damaged spectra, respectively. For each resonance-centered comparison, the frequency window is taken as $f \in [f_r - 2\ \mathrm{kHz},\ f_r + 2\ \mathrm{kHz}]$, where f_r is the corresponding EMI resonance frequency, and the sweep increment is $\Delta f = 0.1$ Hz.

The RMSD values shown in Figs. 14(c)–(f) are further quantify the spectrum variation using the normalized RMSD, both without and with negative-capacitance compensation. The NC-enhanced case amplifies the spectrum-level damage signature relative to the conventional EMI configuration, confirming its effectiveness as a traditional enhancement strategy. However, the response remains distributed over multiple resonances and still relies on spectrum comparison over a frequency band, rather than producing a single concentrated indicator.

To characterize the spectrum-level variation in the proposed NHT-enhanced EMI method, we apply the same RMSD evaluation on the voltage response spectrum. Fig. 15(a) summarizes the normalized RMSD

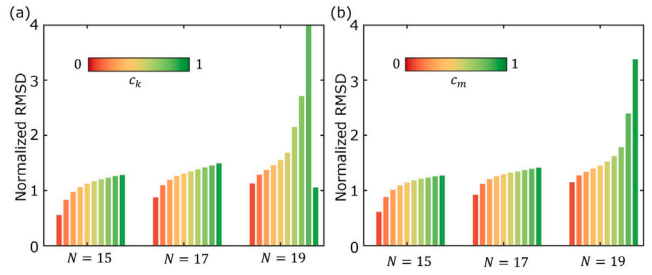


Fig. 15. Normalized RMSD from the NHT-enhanced EMI method. (a) Normalized RMSD of the variation in the voltage response spectrum under different stiffness perturbations, the calculation is the same as the conventional cases. (b) Normalized RMSD by variation of voltage response spectrum under mass perturbations.

of the corresponding voltage spectra under stiffness perturbation for different circuit sizes. Here the RMSD is computed using the same procedure and parameter settings as in the conventional EMI case, with the only change being that the evaluation window is centered at the topological frequency f_t. The NHT-enhanced configuration yields a stronger RMSD response than the conventional EMI method, reflecting the amplified spectral variation induced by the boundary-sensitive topological mode. However, this RMSD comparison is not the main objective of the proposed method; instead, the central sensing observable is the topological frequency shift δf. Fig. 15(b) summarizes the corresponding normalized RMSD. Using the same evaluation procedure and parameter settings as in the conventional EMI case, with the frequency window centered at f_t, the NHT-enhanced configuration again exhibits a stronger RMSD response than the conventional method.

Appendix D. Finite-element implementation and numerical settings

The topological frequency shifts reported in Sections 5–7 are obtained from coupled electromechanical finite-element simulations implemented in COMSOL Multiphysics. In the numerical model, the host structure and bonded piezoelectric patch are described by the linear piezoelectric equations with perfect bonding, while the external NHT circuit is coupled through terminal variables using the Global ODE/DAE interface. This appendix summarizes the finite-element formulation, circuit coupling strategy, mesh-convergence verification, and key numerical settings used in the simulations.

D.1. Coupled electromechanical finite-element formulation

The bonded piezoelectric patch is modeled using the linear stress–charge constitutive relation,

$$\boldsymbol{\sigma} = \boldsymbol{c}^E\boldsymbol{\varepsilon} - \boldsymbol{e}^\top\mathbf{E}, \qquad \mathbf{D} = \boldsymbol{e}\boldsymbol{\varepsilon} + \boldsymbol{\epsilon}\mathbf{E}, \tag{58}$$

with $\mathbf{E} = -\nabla\phi$. After standard finite-element discretization and condensation of the electrode degrees of freedom to a terminal voltage V, the coupled frequency-domain system can be written as

$$\begin{bmatrix} \mathbf{Z}_m(\omega) & \boldsymbol{\Theta} \\ \boldsymbol{\Theta}^\top & -C_p \end{bmatrix} \begin{bmatrix} \mathbf{d} \\ V \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ Q \end{bmatrix}, \tag{59}$$

where $\mathbf{Z}_m(\omega) = \mathbf{K}_{uu} - \omega^2\mathbf{M}$ is the structural dynamic-stiffness matrix, $\boldsymbol{\Theta}$ is the piezoelectric electromechanical coupling vector, and C_p is the effective intrinsic capacitance of the piezoelectric patch. Eq. (59) shows that the finite-element model provides a terminal electromechanical relation between the structural and the electrical variables of the bonded piezoelectric transducer. The piezoelectric patch is modeled as perfectly bonded to the host structure within the linear piezoelectric framework. Different from the continuum host structure, the external

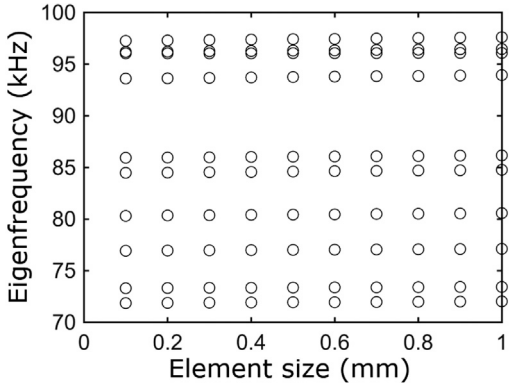


Fig. 16. The convergence analysis of the eigenfrequencies in the finite element beam model with different element sizes.

Table 1
Material properties of structure and piezo patch.

ρ_b	2700 kg/m^3	ρ_p	7600 kg/m^3
c_{11}	126 GPa	e_{31}	−6.55 C/m^2
c_{12}	79.5 GPa	e_{15}	17 C/m^2
c_{13}	84.1 GPa	e_{33}	23.3 C/m^2
c_{33}	117 GPa	ϵ_{11}	1700
c_{44}	23 GPa	ϵ_{33}	1470
c_{66}	23.2 GPa	E_b	70 GPa

Table 2
Circuit parameters in Section 5 and Section 6.

C_1	1.2 nF	C_2	1.5 nF
C_3	0.5 nF	L	0.001 H
N	31, 33, 35		

Table 3
The location parameters of piezoelectric patches in plate setups.

No.	1	2	3	4	5	6	7	8	9	10	11	12
x_s^n(cm)	3	7	11	15	3	7	11	15	3	7	11	15
y_s^n(cm)	3	3	3	3	9	9	9	9	15	15	15	15

NHT circuit is represented as a finite-dimensional lumped network governed by circuit equations. Accordingly, in COMSOL the external circuit is implemented through the Global ODE/DAE interface [78,86], which serves as a numerical co-simulation strategy to enforce the two-terminal electromechanical coupling between the piezoelectric transducer and the circuit. The charge flowing through the piezoelectric electrodes is obtained by integrating the electric displacement over the electrode surfaces. We emphasize that this ODE/DAE implementation is part of the simulation framework used to establish the coupled sensing mechanism, rather than a demonstration of a hardware-integrated or field-deployable realization.

Eq. (59) is the homogeneous coupled eigenvalue problem solved in the COMSOL eigenfrequency study. The corresponding eigenvalues give the natural frequencies of the electromechanically coupled NHT system, and the tracked topological branch is used to quantify the damage-induced frequency shift reported in the manuscript. The detailed material and circuit parameters are listed in Tables 1 and 2.

The location of 9 piezoelectric patches (x_s^n, y_s^n) on the plate is listed in Table 3. The mesh convergence analysis of the beam model is presented in Fig. 16. Since the calculated the eigenfrequencies vary only marginally with mesh refinement, a quadrilateral mesh with an element size of 0.2 mm is selected. The same mesh size is then adopted for the plate model.

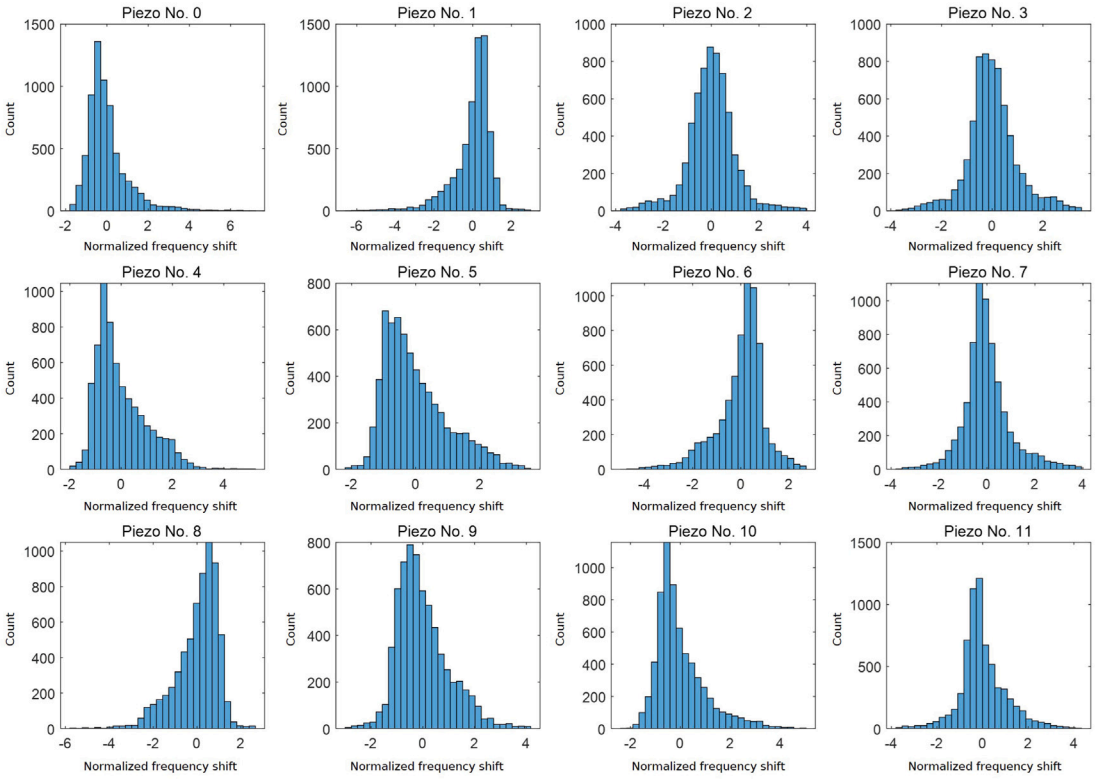


Fig. 17. Statistical distributions of the input features in the plate case dataset. The histograms show the range and variability of the normalized topological frequency shift features used for machine-learning-based damage localization.

Appendix E. Statistical characteristics of the machine-learning dataset

Fig. 17 shows the statistical distributions of the input features for the plate-case dataset. The histograms indicate that the samples span a sufficiently broad range of feature values, without collapsing into a narrow cluster or a degenerate region. Although the spread differs from one feature channel to another, each channel retains observable variability, which supports the adequacy of the generated dataset for supervised training. This observation proves the statistical distributions were introduced to demonstrate that the training data cover a broad and structured parameter space rather than an overly concentrated subset.

Fig. 18(a) further reports the feature importance evaluated by an auxiliary linear regression surrogate, where the absolute coefficient magnitude is used as a first-order measure of the relative influence of each feature on the predicted outputs. The results show that the predictive mapping is not uniformly distributed across all inputs. Instead, several features make stronger contributions, while others play comparatively secondary roles. This behavior suggests that the learned input–output relationship is structured and physically meaningful rather than dominated by random correlations. Since the importance is extracted from a linear surrogate, it should be interpreted as an auxiliary interpretability analysis rather than as an exact internal attribution of the MLP. To evaluate data efficiency, we also trained the MLP using different fractions of the full dataset, from 10% to 100%. As shown in Fig. 18(b), the coefficient of determination R^2 increases rapidly at small training fractions and then approaches saturation. In particular, the model performance rises from $R^2 = 0.81$ at 10% of the training data to about $R^2 = 0.99$ at 30%, and remains close to unity thereafter. This trend indicates that the proposed MLP model is reasonably data-efficient and that its predictive performance becomes stable once a moderate number of training samples is available.

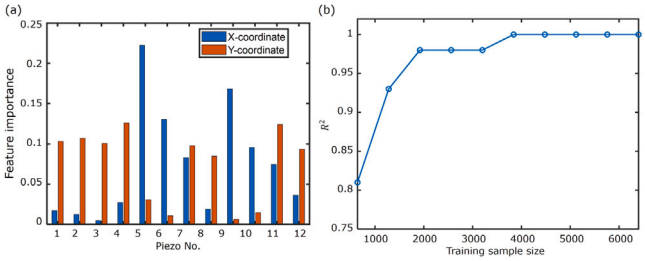


Fig. 18. Feature importance and size effect of the dataset. (a) Feature importance is evaluated using an auxiliary linear regression surrogate, where the absolute coefficient magnitude is used to quantify the relative influence of each feature on the predicted outputs. (b) Model performance under different training sample sizes, reported in terms of the coefficient of determination R^2.

Appendix F. Noise, robustness, and practical considerations

The main text establishes the sensing mechanism of the proposed non-Hermitian topological enhanced EMI framework under an idealized electromechanical model and demonstrates its sensitivity by perturbation analysis and numerical simulations. In practical implementations, however, the observed frequency shift can also be influenced by non-ideal factors such as environmental fluctuations, parasitic circuit elements, component tolerances, active-element limitations, imperfect piezo-structure coupling, and measurement noise. The purpose of this appendix is therefore to examine how these factors enter the proposed framework, to clarify the practical validity range of the present theory, and to evaluate whether the damage-induced topological frequency shift remains distinguishable under representative uncertainty.

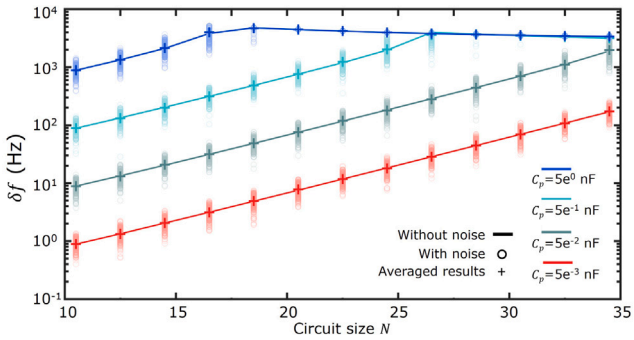


Fig. 19. Noise-robust, size-enhanced frequency shift readout of the NHT circuit under boundary-capacitance perturbations. The topological-mode frequency shift δf is plotted versus circuit size N on a logarithmic scale for four boundary capacitances ($C_p = 5\times10^{0}, 5\times10^{-1}, 5\times10^{-2}, 5\times10^{-3}$ nF). Solid lines represent the noise-free predictions from the circuit eigenvalue problem.

F.1. Stochastic boundary perturbation model

To account for practical uncertainty, we consider the uncertain boundary perturbation into the NHT circuit model in Section 3. We decompose the effective boundary perturbation into a damage-induced part and a noise-induced part:

$$\Delta\mathbf{P}(\omega) = \Delta\mathbf{P}_d(\omega) + \Delta\mathbf{P}_n(\omega), \tag{60}$$

where $\Delta\mathbf{P}_d$ denotes the deterministic perturbation caused by structural damage, and $\Delta\mathbf{P}_n$ represents measurement noise, environmental fluctuation, and other non-ideal effects. In this way, the same boundary perturbation framework used in the main text can be extended to assess practical robustness.

In the following analysis, $\Delta\mathbf{P}_n$ is introduced as a random fluctuation of the effective boundary capacitance (or equivalently the extracted boundary admittance), and the NHT eigenproblem is solved repeatedly to evaluate the resulting spread of the topological frequency shift. Fig. 19 shows representative noisy realizations together with the ensemble-averaged response. The solid lines denote the no-noise prediction from the circuit eigenproblem. Open circles show representative noisy realizations obtained by adding random fluctuations to the boundary perturbation C_p (noise level is $\Delta\mathbf{P}_n \in [-0.2\Delta\mathbf{P}_d, 0.2\Delta\mathbf{P}_d]$). The + markers indicate the ensemble-averaged δf at each N, illustrating that the size-dependent amplification trend persists in the presence of boundary-measurement uncertainty. The noise analysis in Fig. 19 indicates that the size-dependent amplification trend remains observable under moderate uncertainty, but a complete experimental assessment will still require circuit-level stability analysis, tolerance characterization, and repeated measurements under realistic operating conditions.

F.2. Effect of reducing the number of piezoelectric patches

In practical applications, the number of piezoelectric patches may be limited by space, cost, or instrumentation constraints. Therefore, we further examine how the localization performance changes when the number of sensing patches is reduced.

Fig. 20 shows the predicted and true damage coordinates for 12, 9, 6, and 3 piezoelectric patches. The results indicate that the proposed method remains accurate for 12, 9, and 6 patches, with $R^2 = 1.00$, 0.98, and 0.97, respectively. When the number of patches is further reduced to 3, however, the prediction accuracy drops significantly to $R^2 = 0.44$. This result shows that the proposed framework is robust to a moderate reduction in sensor number, but an excessively sparse sensing layout cannot provide sufficient spatial information for reliable damage localization.

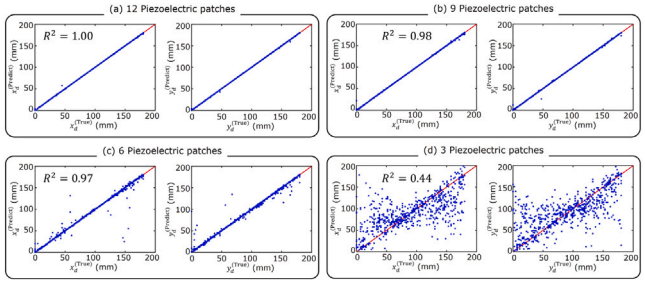


Fig. 20. Comparison between true and predicted damage coordinates for different piezoelectric sensor layout densities: (a) 12 patches, (b) 9 patches (piezoelectric patch Nos. [1 2 4 5 7 8 9 10 11 12]), (c) 6 patches (piezoelectric patch Nos. [1 4 7 9 10]), and (d) 3 patches (Piezoelectric patch No. [1 4 7]). The red diagonal indicates perfect prediction.

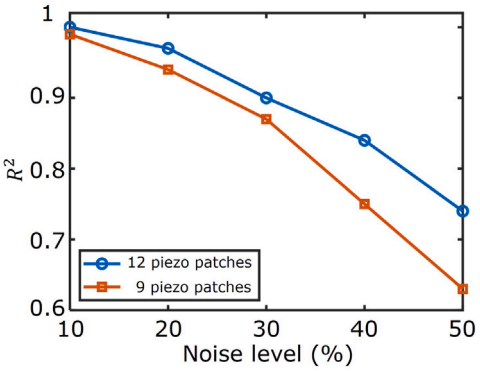


Fig. 21. Coefficient of R^2 for damage-localization prediction as a function of noise level for 12- and 9-piezoelectric-patch layouts.

F.3. Noise effect in data-driven MLP method

The near-perfect prediction results in the main text mainly reflect that the dataset was generated from deterministic FE simulations without measurement noise or environmental uncertainty. To further assess robustness, we added random perturbations to the shifted topological frequency features and repeated the localization analysis. The noisy input is defined as

$$\delta f_i^{(n)} = \delta f_i(1 + \eta\xi_i), \tag{61}$$

where η denotes the prescribed noise level and ξ_i is a random variable. Noise levels of 10%, 20%, 30%, 40%, and 50% were considered.

Fig. 21 shows the coefficient of determination R^2 for the 12- and 9-patch layouts under different noise levels. For the 12-patch case, the corresponding R^2 values are 1.00, 0.97, 0.90, 0.84, and 0.74. For the 9-patch case, the values are 0.99, 0.94, 0.87, 0.75, and 0.63. The prediction accuracy decreases gradually as the noise level increases, rather than collapsing abruptly, which indicates that the proposed framework retains useful predictive capability under moderate uncertainty.

Fig. 22 further compares the true and predicted damage coordinates for representative cases with 20% and 50% noise. At 20% noise, most prediction points remain close to the diagonal, indicating that the localization is still accurate under moderate perturbation. At 50% noise, the scatter becomes more pronounced, but the overall trend is still preserved, especially for the denser 12-patch layout. These results indicate that higher sensor density provides stronger redundancy and therefore better robustness against noisy inputs.

The same MLP architecture and data-processing procedure were used, with only the input frequency shift features being perturbed. These additional tests do not change the deterministic nature of the original FE dataset, but they do provide a direct evaluation of robustness against input uncertainty.

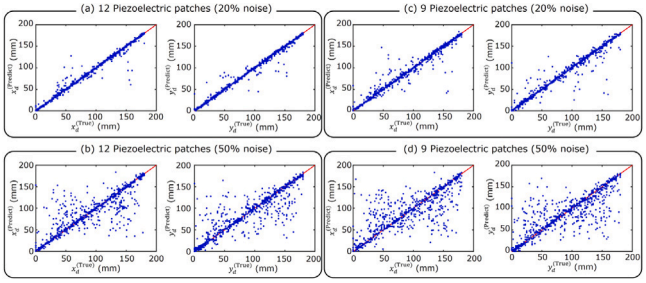


Fig. 22. Comparison between true and predicted damage coordinates under noisy conditions: (a) 12 piezoelectric patches with 20% noise, (b) 12 piezoelectric patches with 50% noise, (c) 9 piezoelectric patches with 20% noise, and (d) 9 piezoelectric patches with 50% noise.

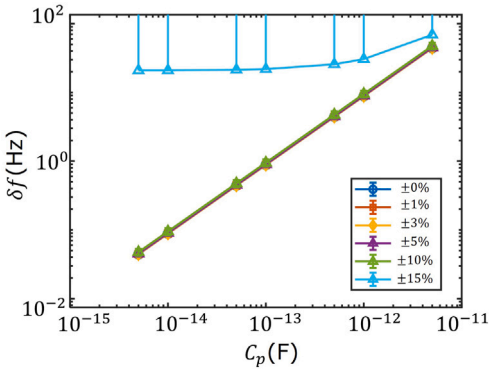


Fig. 23. Robustness of the damage-induced frequency shift ($\delta f = |f_t^{(d)} - f_t^{(0)}|$) response at a fixed circuit scale under component tolerance and parameter drift.

Appendix G. Robust sensing against the disorder in circuit manufacture

To examine robustness against circuit non-idealities at a fixed circuit scale, we perturb the nominal parameters as $C \to C(1+\xi)$, $0 \to R_0(\xi_L)$, and $L \to L(1+\xi)$, where the deviation factors represent tolerance and drift by $\xi \in [0, 15\%]$. For each perturbed circuit, we compute both the intact frequency $f_t^{(0)}$ and the frequency after applying the same boundary perturbation, and define the sensing output by $\delta f = |f_t^{(d)} - f_t^{(0)}|$. The results show in Fig. 23, under moderate parameter variation ($\xi \in [0, 10\%]$), the damage-induced differential shift remains observable after recalibration of the intact state, indicating that the sensing function is preserved even though the nominal operating point drifts. However, the frequency shifts fails at when the introducing is large, when $\xi = 15\%$.

Appendix H. Practical design of the balancing impedance/admittance

The NHT-enhanced EMI method requires a balancing branch to remove the intact electromechanical boundary contribution of the piezoelectric patch and host structure, so that the topological mode remains within its sensitive perturbation range. In the present formulation, the natural calibration quantity is the boundary admittance rather than the boundary impedance. For the intact structure, the piezoelectric patch and host together contribute an effective terminal admittance

$$Y_{\text{piezo},0}(\omega) = i\omega \left(C_p^T + \theta^2 z_{m,0}(\omega) \right), \tag{62}$$

where C_p^T is the intrinsic piezoelectric capacitance and

$$z_{m,0}(\omega) = \mathbf{b}_m^\top \mathbf{Z}_{m,0}^{-1}(\omega) \mathbf{b}_m \tag{63}$$

is the projected intact mechanical compliance seen by the piezoelectric element. Therefore, the ideal balancing target is

$$Y_b^{\text{ideal}}(\omega) = -Y_{\text{piezo},0}(\omega). \tag{64}$$

At the designed topological frequency ω_t, this choice cancels the intact boundary term and recovers the intended unperturbed NHT eigenvalue problem.

Equation above shows that the ideal balancing branch contains two parts: a negative-capacitive part that compensates the static piezoelectric capacitance, and a frequency-dependent correction associated with the intact host structure. However, in a multimodal/continuous structure, Y_b^{ideal} is generally a high-order complex function of frequency, so exact broadband synthesis is difficult in experiment. However, the proposed sensing method uses only the frequency shift of a topological mode in a narrow neighborhood of ω_t. Consequently, Y_b need not reproduce the intact electromechanical admittance over the full spectrum; it only needs to match the target locally near the operating frequency.

This observation motivates a hierarchy of practical approximations. The lowest-order design retains only the static term,

$$Y_b^{(0)}(\omega) = -i\omega(C_p^T + \theta^2 z_{m,0}(\omega_t)), \tag{65}$$

which corresponds to a negative capacitor and already removes the dominant capacitive offset. A refined narrow-band design includes local frequency dependence of the intact structural term by a Taylor expansion about ω_t,

$$Y_b^{(1)}(\omega) = Y_b^{(0)}(\omega) + (\omega - \omega_t) \left[\partial Y_b^{\text{ideal}} / \partial \omega \right], \tag{66}$$

or

$$Y_b^{(2)}(\omega) = Y_b^{(1)}(\omega) + (\omega - \omega_t)^2 \left[\partial^2 Y_b^{\text{ideal}} / \partial \omega^2 \right]. \tag{67}$$

More generally, one may determine a weighted local polynomial fit over a finite frequency window around ω_t, which provides a convenient numerical surrogate for the narrow-band target. Fig. 24 shows how the balancing admittance design affects the NHT sensing response. Fig. 24(a) compares the designed admittance with the ideal target near the topological frequency f_t. The first-order design improves the local cancellation at f_t, but its accuracy deteriorates away from the operating point, whereas the second-order design remains nearly indistinguishable from the ideal target over the narrow band of interest. As shown in Figs. 24(b) and 24(c), this mismatch directly influences both the intact and damaged voltage-response spectra. Lower-order designs distort the spectral evolution and reduce the separation between the intact and damaged peaks, leading to lower shift sensitivity. By contrast, the second-order design reproduces the ideal spectral response in both cases. This trend is further confirmed in Fig. 24(d), where the second-order design closely follows the ideal damage-induced frequency shift δf, while the zeroth- and first-order designs systematically underestimate it.

For the FE model, the balancing admittance can be identified from the intact system using the same structural model and piezoelectric configuration as in the conventional EMI analysis. In particular, the intact terminal admittance $Y_0(\omega)$ can be obtained by the blue curve in Fig. 9(b), which has the same mechanical and piezoelectric part.

In practice, the balancing impedance/admittance can be decomposed into a negative-capacitance part and a higher-order correction part. The negative-capacitance component is introduced primarily to compensate the intrinsic piezoelectric capacitance C_p^T, and can be realized using a current-feedback operational-amplifier-based floating negative impedance circuit [87]. The remaining frequency-dependent terms are used to match the intact electromechanical admittance locally in the neighborhood of the target topological frequency. In this sense, the design should be interpreted as a local approximation target rather than as a literal passive lumped circuit identity over the full spectrum. Its realization can therefore rely on established active synthetic-impedance techniques: op-amp-based negative-capacitance circuits for

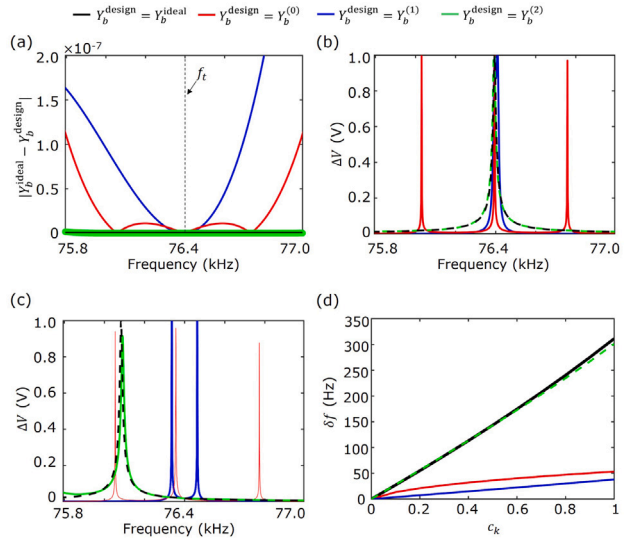


Fig. 24. Effect of balancing-admittance design on the NHT sensing response. (a) Magnitude of the mismatch between the ideal balancing admittance Y_b^{ideal} and the designed admittance Y_b^{design} in the vicinity of the target topological frequency f_t, comparing the zeroth-, first-, and second-order designs. (b) Voltage-response spectrum of the intact system under different balancing-admittance designs. The circuit size is $N = 17$. (c) Voltage-response spectrum of the damaged system under the same designs when $c_k = 1$. (d) Damage-induced topological frequency shift δf versus normalized stiffness perturbation c_k.

the leading capacitive compensation, together with analog or digital synthetic-impedance circuits for the higher-order correction terms [88,89]. A new Z_b (or equivalently Y_b) setup is generally required only when the measurement configuration changes, for example when a different piezoelectric location is used or a different operating frequency is selected, since both alter the intact projected electromechanical admittance $z_{m,0}$. Accordingly, the practical procedure is to determine Z_b once for each sensor-location/operating-frequency pair under the reference undamaged condition, after which the same setting can generally be used for extended periods. Re-adjustment is needed only when environmental drift, component aging, or hardware reconfiguration causes a significant change in the local electromechanical baseline.

Appendix I. Experimental guideline for future validation

Although experimental validation is beyond the scope of the current study, the proposed framework can be examined using a piezo-structure-circuit platform. A practical validation route is to use a beam or plate specimen instrumented with a bonded piezoelectric patch. The intact piezo-structure system should first be characterized through its baseline impedance/admittance spectrum, which provides the reference electromechanical response and the information needed for the balance impedance design discussed in this Appendix H. The external NHT circuit can then be implemented as a lumped electrical network connected to the two piezoelectric terminals. In practice, the nonreciprocal couplings may be synthesized using op-amp-assisted directional circuit elements. Since practical realization of such active circuits is affected by finite bandwidth, parasitic loading, and electrical noise, additional stabilization measures are generally required, such as power-rail decoupling, compensation of input capacitance, and output isolation resistor to mitigate capacitive loading [60,62,90]. These non-idealities do not change the proposed sensing principle, but they should be accounted for in a hardware implementation. After the intact structural measurement and balance impedance is established, controlled structural perturbations can be introduced, for example through local mass addition or equivalent stiffness reduction. For each damage state, frequency sweeps are performed to record the voltage or impedance response of the coupled system. In the proposed NHT-enhanced EMI framework, the main observable is the shift of the topological-mode frequency, which can be directly compared with the theoretical and numerical predictions as a function of damage level, circuit size, and operating frequency.

Data availability

Data will be made available on request.

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